DOMINANT PERMUTATIONS, POSITIVE MIXED VARIABLE FORMULAS, AND PRODUCT–SPLITTING EQUIVALENCE FOR (DOUBLE) SCHUBERT POLYNOMIALS

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ABSTRACT. We present new positive combinatorial formulas for triple Schubert coefficients in mixed-variable products of double Schubert polynomials, $\mathfrak{S}_u(x;y)\,\mathfrak{S}_v(x;z)$, expressed in the $\mathfrak{S}_w(x;y)$ basis, in the case where u is dominant. This result complements earlier results where v was assumed dominant. To generalize further, we introduce a relation on permutations that we call "domination," under which dominant permutations dominate all others. When u dominates w, we obtain an explicit positive formula for $c^w_{u,v}(y;z)$, which in particular yields positivity when u and v have separated descents.

In addition to these new positivity results, we establish a product–splitting equivalence for double Schubert polynomials, relating triple Schubert coefficients to splitting coefficients. This equivalence highlights structural symmetries and provides reduction methods for computing coefficients that are otherwise difficult to access. Taken together, these results extend the range of double Schubert polynomial products for which positivity can be established combinatorially and yield explicit formulas that can be implemented computationally.

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INTRODUCTION

Double Schubert polynomials $\mathfrak{S}_u(x;y)$, indexed by permutations, constitute a central basis in Schubert calculus. Their multiplication with respect to different sets of coefficient variables defines triple Schubert structure constants, also called triple Schubert coefficients, via the relation

$$\mathfrak{S}_u(x;y)\ \mathfrak{S}_v(x;z)\ =\ \sum_w c^w_{u,v}(y;z)\ \mathfrak{S}_w(x;y).$$

The central objective in this area is to obtain explicit combinatorial formulas that exhibit positivity, in analogy with the Littlewood-Richardson rule for Schur polynomials. Since the triple structure constants are

2020 *Mathematics Subject Classification.* 05E05, 05A05, 05A15, 14N10.

Key words and phrases. Double Schubert polynomial, Schubert polynomial, triple Schubert coefficient, triple structure constant, splitting coefficients, Pieri.

polynomials, their positivity is understood in terms of suitable differences of coefficient variables, rather than ordinary monomials in y and z.

Gao and Xiong proved the positivity conjecture of [22, Conjecture 1.1] in their FPSAC 2026 paper *Positivity in triple Schubert calculus* [9, Theorem 1.1]. With our ordering of the factors and coefficient variables, their theorem states that every triple Schubert coefficient $c^w_{u,v}(y;z)$ is a polynomial in the differences $y_i - z_j$ with nonnegative integer coefficients. Their geometric proof uses a refinement of Graham's positivity theorem [10], but does not provide a general positive combinatorial formula. Positivity in arbitrary differences $y_i - z_j$ need not make Graham positivity in the simple roots $y_{i+1} - y_i$ manifest after specializing $z = y$.

In this paper, we give explicit positive combinatorial formulas in broad new cases, complementing Gao and Xiong's geometric positivity theorem. We introduce a relation on permutations that we call "domination," extending the classical notion of dominant permutation. We prove that if u dominates w, then the coefficient $c^w_{u,v}(y;z)$ has a positive combinatorial formula. This complements the earlier formulas of [22] for dominant v and yields positive combinatorial formulas when u and v have separated descents in either orientation.

We also establish an equivalence between products and splitting coefficients of double Schubert polynomials. This equivalence enables us to compute products from splitting coefficients and conversely, thereby exposing structural symmetries and providing practical methods to reduce complex cases to simpler ones. The result extends earlier work of Bergeron and Sottile on ordinary Schubert polynomials and gives a framework for organizing the difficulty of multiplication problems in terms of splitting-coefficient computations.

Software used. All computation in this work was carried out using schubmult `schubmult` [23], a free Python package developed by the author. The package is available on PyPI and comes with command-line executables. It supports:

- Products of ordinary Schubert polynomials
- Products of double Schubert polynomials (both same-variable and mixed-variable)
- Splitting coefficients of Schubert polynomials
- Products of quantum/quantum double Schubert polynomials (also same or mixed variable, including parabolic)
- Conjectural equivalents for Grothendieck/double Grothendieck/quantum double Grothendieck polynomials

Questions, bug reports, or other inquiries about the package can be directed to schubmult@gmail.com.

1. The domination formula

1.1. Statement of the formula with examples.

Definition 1.1.1 (S_∞ as a Coxeter group, window notation)**.** For standard results and notation regarding Coxeter groups, we refer the reader to [4] or [11]. S_∞ will refer to the infinite symmetric group of permutations $\mathbb{N} \to \mathbb{N}$ fixing all but finitely many elements, and S_n is the subgroup of all $u \in S_\infty$ such that $u(i) = i$ for all $i > n$. We will use window notation, where we express a permutation u as a permutation of the first n positive integers (enclosed by brackets []) such that if we write

$$u = [a_1, a_2, \ldots, a_n]$$

where $a_i \neq a_j$ if $i \neq j$ and $1 \leq a_i \leq n$ for all i, then $u(i) = a_i$ for $1 \leq i \leq n$, and $u(i) = i$ if $i > n$. An alternative notation, code notation $[\![\cdots]\!]$, will be defined in Definition 1.1.5.

We use the standard notation of s_i as the adjacent transposition such that right multiplication exchanges the elements at indices i and $i+1$, and t_{ij} is the transposition such that right multiplication exchanges the elements at indices i and j. These are the simple reflections (resp. reflections) in a Coxeter presentation of S_∞.

Definition 1.1.2 (Separated descents)**.** Two permutations u and v are said to have *separated descents* if there exists a positive integer p such that $\ell(us_i) > \ell(u)$ for all $i < p$ and $\ell(vs_j) > \ell(v)$ for all $j > p$. It is important to note that this definition is not symmetric in u and v. [22] and [6] address the separated descents case of positivity of $c^w_{u,v}(y;z)$ as defined above. Positivity of $c^w_{v,u}(y;z)$ is proved in this article.

Definition 1.1.3 (Special group elements $c[p,q]$, $d[p,q]$, $r[p,q]$, the homomorphism σ_p)**.** For $p \geq 1$, $k > 0$, define the cycle

$$c[p,k] = s_{k+1-p} s_{k+2-p} \cdots s_k$$

and also define

$$r[p,k] = s_{k+p-1}s_{k+p-2}\cdots s_k$$

We use the same notation for these as in [24], and $c[p,k]$ corresponds to elementary symmetric polynomials, while $r[p,k]$ corresponds to complete symmetric polynomials. The choices of "c" refer to the word "column" and "r" to "row," referring to the shape of the indexing partition of the corresponding Schur polynomial.

For $q \geq 0$ define

$$d[p,q] = s_p s_{p+1}\cdots s_{p+q-1}$$

(where by convention $d[p,0] = 1$) and define

$$d[q] = d[1,q]$$

Note that $d[p,q] = c[q,p+q-1]$, so that this is merely a convenient way to re-express the c permutations. If $q_1 \leq q_2$, then $d[p,q_1]^{-1}d[p,q_2] = d[p+q_1, q_2-q_1]$.

Define a partial homomorphism σ_p for a not necessarily positive integer p by $\sigma_p(s_i) = s_{i+p}$ whenever $i+p > 0$. This is defined for all w such that $w = s_{i_1}\cdots s_{i_m}$ with $i_j + p > 0$ for all j. σ_1 will be used often, so σ with no subscript will indicate σ_1 to avoid clutter.

Definition 1.1.4 (Sottile's Pieri relations $\xrightarrow{k}, \overset{k}{\Longrightarrow}$)**.** Following [24], with slightly different notation, for permutations u, w we declare that $u \xrightarrow{k} w$ if there exist transpositions $t_{a_1b_1}, \ldots, t_{a_pb_p}$ for some p with $0 \leq p \leq k$ such that:

(1) $a_i \leq k < b_i$ for all i.
(2) $a_i \neq a_j$ if $i \neq j$.
(3) $\ell(ut_{a_1b_1}\cdots t_{a_ib_i}) = \ell(u) + i$ for all $1 \leq i \leq p$.
(4) $w = ut_{a_1b_1}\cdots t_{a_pb_p}$.

This definition is used in Sottile's Pieri formula in the reference and is central to the development of our formula. We also define a relation $u \overset{k}{\Longrightarrow} w$, also defined by Sottile, if there exist transpositions $t_{a_1b_1}, \ldots, t_{a_pb_p}$ for some $p \geq 0$ such that (1), (3), and (4) hold, but instead of (2) we have that $b_i \neq b_j$ if $i \neq j$. Note that, unlike for $\xrightarrow{k}$, p can be greater than k with the relation still holding. After introducing the Pieri relations, which explain how permutations can be changed using carefully crafted transposition sequences, we go on to the code of a permutation, another essential combinatorial tool. This will enable us to compactly depict permutations in a manner that organically engages with the previously established relations.

Definition 1.1.5 ($\mathfrak{c}(w)$, $\mathfrak{c}^*(w)$, dominant permutations, the dominant approximation $\mathfrak{d}(v)$, code representation $[\![\cdots]\!]$)**.** Given $w \in S_\infty$, the *code* $\mathfrak{c}(w)$ (also known as the Lehmer code) of w as follows:

$$\mathfrak{c}_i(w) = \#\{j > i \mid w(i) > w(j)\}$$

In other words, $\mathfrak{c}_i(w)$ is the number of inversions (i,j) of w, with i fixed. We also define the dual code

$$\mathfrak{c}^*(w) = \mathfrak{c}(w^{-1})$$

which is simply the code of the inverse. The concept of permutation is quite ancient, found for instance in [16].

Some useful facts about the code, which we will leave to the reader, are as follows: $u(i) < u(i+1)$ if and only if $\mathfrak{c}_i(u) \leq \mathfrak{c}_{i+1}(u)$, and in that case

$$\mathfrak{c}_j(us_i) = \begin{cases} \mathfrak{c}_j(u) & \text{if } j \notin \{i, i+1\} \\ \mathfrak{c}_{i+1}(u) + 1 & \text{if } j = i \\ \mathfrak{c}_i(u) & \text{if } j = i+1 \end{cases}$$

Furthermore, $\mathfrak{c}_i^*(w)$ is equal to the number of inversions (i,j) of w^{-1}, which is to say that $w^{-1}(i) > w^{-1}(j)$. This means that, in terms of w, if q is the index such that $w(q) = i$, then $\mathfrak{c}_i^*(w)$ is the number of indices $p < q$ such that $w(p) > i$. It follows easily that.

$$w(\mathfrak{c}_1^*(w) + 1) = 1$$

The inverse function of the code will be denoted by $[\![\cdots]\!]$, meaning $[\![\mathfrak{c}(w)]\!] = w$. For example, since

$$\mathfrak{c}([1,3,5,2,4,7,6]) = (0,1,2,0,0,1)$$

we have

$$[\![0,1,2,0,0,1]\!] = [1,3,5,2,4,7,6]$$

A permutation μ is said to be a *dominant permutation* if $\mathfrak{c}(\mu)$ is a partition. Dominant permutations can also be characterized as the permutations that avoid the pattern 132, that is to say, there are no indices $i < j < k$ such that $u(i) < u(k) < u(j)$ [20].

For a permutation v, we define a dominant permutation $\mathfrak{d}(v)$ following [22], recursively as follows. If v is dominant, then $\mathfrak{d}(v) = v$. Otherwise, let i be the maximal index such that $\mathfrak{c}_i(v) < \mathfrak{c}_{i+1}(v)$; then set

$$\mathfrak{d}(v) = \mathfrak{d}(vs_i)$$

This recursive process always terminates at a unique dominant permutation, since at each step there is only one possible choice of index.

Definition 1.1.6 ($\varphi_{i,n}$, $\mathcal{D}_i(v)$, $Q_i(v',v)$)**.** For a permutation v' such that $v' \in S_n$, define

$$\varphi_{i,n}(v')(j) = \begin{cases} v'(j) & \text{if } j < i \\ n+1 & \text{if } j = i \\ v'(j-1) & \text{if } i < j \leq n+1 \\ j & \text{if } j > n+1 \end{cases}$$

Now fix $v \in S_n$ and $i \geq 1$ an integer, we define a relation $\overset{i}{\searrow}$ by declaring that $v \overset{i}{\searrow} v'$ if

$$v \overset{i}{\Rightarrow} \varphi_{i,n}(v')$$

Equivalently, $v \overset{i}{\searrow} v'$ if whenever $v \in S_n$ and n is minimal, we have that v satisfies the relation $\overset{i}{\Rightarrow}$ with respect to the permutation obtained from v' by inserting $n+1$ at position i. We note that this concept was introduced by Bergeron and Sottile in [3].

If $v \overset{i}{\searrow} v'$, we define a set of integers $Q_i(v',v)$ by

$$Q_i(v',v) = \{v(j) \mid j > i \text{ and } v'(j-1) = v(j)\}$$

Given the fact that any element of S_∞ fixes all but finitely many positive integers, it follows that $Q_i(v',v)$ is a finite set.

We then define $\mathcal{D}_i(v)$ to be all permutations $v' \in S_\infty$ such that $v \overset{i}{\searrow} v'$. The permutations $\mathcal{D}_i(v)$ arise when pulling a variable out of a Schubert polynomial and expressing the coefficients of the powers of this variable as Schubert polynomials in the remaining variables, as is done in [2], from which this definition essentially comes. The associated set $Q_i(v',v)$ is specific to the case of double Schubert polynomials.

Example 1.1.7. The permutation $\mu = [5,4,6,3,1,2]$ has code $(4,3,3,2)$ and hence is dominant. We then have that $\mathfrak{c}^*(\mu) = (4,4,3,1)$.

Let $v = [4,3,1,5,2]$. Then $\mathfrak{c}(v) = (3,2,0,1)$, hence $\mathfrak{d}(v) = vs_3 = [4,3,5,1,2]$. We have that $\mathfrak{c}(\mathfrak{d}(v)) = (3,2,2)$ and $\mathfrak{c}^*(\mathfrak{d}(v)) = (3,3,1)$. We have that $v^{-1} = [3,5,2,1,4]$, $\mathfrak{c}(v^{-1}) = (2,3,1)$, hence $\mathfrak{d}(v^{-1}) = v^{-1}s_1 = [5,3,2,1,4]$. $\mathfrak{c}(\mathfrak{d}(v^{-1})) = (4,2,1)$ and $\mathfrak{c}^*(\mathfrak{d}(v^{-1})) = (3,2,1,1)$. We have that $v \overset{3}{\searrow} [4,3,1,2]$ and $v \overset{3}{\searrow} [4,3,2,1]$, and this list is exhaustive. We then have that

$$Q_3([4,3,1,2],v) = \{2\}$$

and $Q_3([4,3,2,1],v) = \emptyset$.

Definition 1.1.8 (ϕ, ϕ_m, *k-dominates*, *dominates*)**.** A permutation u is said to 1-*dominate* the permutation w if for each index j with $\mathfrak{c}_1^*(u) < j \leq \mathfrak{c}_1^*(w)$ we have $u(j) < u(j+1)$. Equivalently, u has no descents strictly to the right of position $\mathfrak{c}_1^*(u)$ up to position $\mathfrak{c}_1^*(w)$.

For $i \geq 0$, we define a function $\phi_i : S_\infty \to S_\infty$ by declaring that

$$\mathfrak{c}_j^*(\phi_i(u)) = \mathfrak{c}_{j+i}^*(u)$$

for all j, essentially "chopping off" the first i elements of the dual code. We recursively say that u *k-dominates* w for $k > 1$ if u $(k-1)$-dominates w and $\phi_{k-1}(u)$ 1-dominates $\phi_{k-1}(w)$. We then say that u *dominates* w if u k-dominates w for all $k \geq 1$. For simplification of notation, we write

$$\phi = \phi_1$$

Example 1.1.9. The permutation $u = [2, 3, 5, 1, 4]$ dominates the permutation $w = [2, 6, 3, 4, 7, 1, 5]$ but is not dominant. To see that u dominates w, note that $\mathfrak{c}_1^*(u) = 3$, $\mathfrak{c}_1^*(w) = 5$, and $u(4) < u(5) < u(6)$, so u 1-dominates w. We have that $\phi(u) = [1, 2, 4, 3]$ and $\phi(w) = [1, 5, 2, 3, 6, 4]$. Clearly $\phi(u)$ 1-dominates $\phi(w)$ since $\mathfrak{c}_1^*(\phi(u)) = \mathfrak{c}_1^*(\phi(w))$. We have that $\phi_2(u) = [1, 3, 2]$ and $\phi_2(w) = [4, 1, 2, 5, 3]$. Since $\mathfrak{c}_1^*(\phi_2(w)) = 1$ and $u(1) < u(2)$, we have that u 3-dominates w. It is indeed true that u dominates w.

Definition 1.1.10 (Double Schubert polynomials $\mathfrak{S}_u(x; y)$, divided difference operators ∂^v, ∂_v^w, $c_{u,v}^w(y; z)$)**.** We denote the double Schubert polynomial indexed by $u \in S_\infty$ in the variables x and y by $\mathfrak{S}_u(x; y)$. We define coefficients $c_{u,v}^w(y; z)$ by

$$\mathfrak{S}_u(x; y)\, \mathfrak{S}_v(x; z) = \sum_{w \in S_\infty} c_{u,v}^w(y; z)\, \mathfrak{S}_w(x; y)$$

Divided difference operators are indexed by an element of S_∞ as a superscript, for example ∂^u. Define skew divided difference operators ∂_v^w by defining for polynomials P, Q

$$\partial^w(PQ) = \sum_{v \in S_\infty} \partial^v(P) \partial_v^w(Q)$$

This differs from the definition of skew divided difference operators in [17], but agrees with that in [21], and in our opinion is more natural. We have that

$$c_{u,v}^w(x; z) = \partial_u^w(\mathfrak{S}_v(x; z))$$

An explanation of this fact can be found in [22].

Definition 1.1.11 (Polynomial variable omission $x^{(i)}$, subscript substitution y_A, index shift function τ)**.** Define

$$x^{(i)} = (x_1, \ldots, \widehat{x_i}, \ldots)$$

For a sequence of multiple indices in the superscript, we omit all of them.

For a set of positive integers A and a sequence of indeterminates such as $y = (y_i)_{i=1}^\infty$, we denote

$$y_A = \{y_a \mid a \in A\}$$

While this is expressed as a set, the implication is that the possibly nonconsecutive indices occur in increasing order.

We define a homomorphism $\tau : \mathbb{Z}[x, y, z] \to \mathbb{Z}[x, y, z]$ by specializing $\tau(x_j) = x_{j+1}$, $\tau(y_j) = y_j$ and $\tau(z_j) = z_j$. We denote the composition of τ with itself i times by τ^i.

Definition 1.1.12 ($H_p(x; y)$, $E_p(x; y)$, $\prod(a \mid B)$)**.** Define

$$H_p(x; y) = \prod_{i=1}^{p} (x_1 - y_i)$$

This is actually a double Schubert polynomial, namely

$$H_p(x; y) = \mathfrak{S}_{r[p,1]}(x; y)$$

Define also

$$E_p(x; y) = \prod_{i=1}^{p} (x_i - y_1)$$

Let $\mathcal{F}$ denote the set of finite subsets of $\mathbb{Z}[x, y, z]$. We define a function

$$\prod : \mathbb{Z}[x, y, z] \times \mathcal{F} \to \mathbb{Z}[x, y, z]$$

that is $H_p(x; y)$ thinly disguised by substitution, by

$$\prod(a \mid B) = \prod_{b \in B} (a - b)$$

Typically, a will be a single indeterminate and B will be a set or sequence of indeterminates, rather than arbitrary elements of the ring. For example,

$$\prod\left(x_1 \mid y_{[p]}\right) = \prod_{i=1}^{p}(x_1 - y_i) = H_p(x;y)$$

$$\prod\left(-y_1 \mid (-x)_{[p]}\right) = \prod_{i=1}^{p}(x_i - y_1) = E_p(x;y)$$

By a mild abuse of notation, we may write such expressions as

$$\prod\left(x_i \mid y_1, y_3, y_4\right) = (x_i - y_1)(x_i - y_3)(x_i - y_4)$$

We may now state our domination formula. For clarity in interpreting the notation, we refer the reader to the index at the end of the article, which cross-references the formal definitions.

Theorem 1.1 (The domination formula)**.** *Let $u, v, w \in S_\infty$ and suppose u dominates w. Then $c^w_{u,v}(y;z) = 0$ unless $\mathfrak{c}^*_j(w) \geq \mathfrak{c}^*_j(u)$ for all j. If that condition is satisfied, let m be the maximal index such that $\mathfrak{c}^*_m(u) \neq 0$. For each $1 \leq k \leq m$ define*

$$d_k = d[\mathfrak{c}^*_k(u) + 1, \mathfrak{c}^*_k(w) - \mathfrak{c}^*_k(u)]$$

Then

$$c^w_{u,v}(y;z) = \sum_{v_1,\ldots,v_m} \left(\prod_{i=1}^{m} \prod\left(y_i \mid z_{A_i}\right)\right) \mathfrak{S}_{v_m\phi_m(w)^{-1}}(\tau^m y; z)$$

where $v_0 = v$, $\ell(v_m\phi_m(w)^{-1}) = \ell(v_m) - \ell(\phi_m(w))$, and v_k ranges over all elements such that

$$v_k \overset{\mathfrak{c}^*_k(u)+1}{\searrow} v_{k-1}d_k^{-1}$$

and

$$\ell(v_{k-1}d_k^{-1}) = \ell(v_{k-1}) - \mathfrak{c}^*_k(w) + \mathfrak{c}^*_k(u)$$

and we define

$$A_k = Q_{\mathfrak{c}^*_k(u)+1}(v_k, v_{k-1}d_k^{-1})$$

for all k.

In particular, this holds for all v and w when u is dominant (see Lemma 1.2.4).

We illustrate the terms in the sum with diagrams, which are conceptually similar to the diagrams in [22]. In this diagram, the relevant permutations are arranged column-wise. Circled entries represent elements of the Q set (Definition 1.1.6). Visually, these correspond to numbers that don't change from column to column.

More specifically, each diagram is a rectangles that contain the permutations v_i, $v_i d_{i+1}^{-1}$, and $v_m\phi_m(w)^{-1}$ in sequence. If $v_i \neq v_i d_{i+1}^{-1}$ then there will be an arrow from v_i to $v_i d_{i+1}^{-1}$ labeled with d_{i+1}. In between $v_{i-1}d_i^{-1}$ and v_i there will be a vertical bar, and a bullet occupying position $\mathfrak{c}^*_i(u) + 1$, with the elements below displaced to the rows below, and all permutations afterwards in the sequence between $v_{i-1}d_i^{-1}$ and the next arrow will have elements displaced around previous bullets as though the row had been removed. If a number below the bullet in some column is equal to the number immediately to the left of the bar in the same row, then this number will be circled, and this contributes a factor of $y_a - z_b$, where a is the region number (counting from 0 beginning at the left, increasing by 1 as we pass each bar) and b is the circled number. If $\phi_m(w) \neq 1$, there will be an arrow between v_m and $v_m\phi_m(w)^{-1}$ labeled with $\phi_m(w)$, otherwise $v_m = v_m\phi_m(w)^{-1}$ will be the final permutation in the sequence. If $v_m\phi_m(w)^{-1} \neq 1$, this contributes a factor of $\mathfrak{S}_{v_m\phi_m(w)^{-1}}(\tau^m y; z)$, and will be written in a bold font in red in a column labeled $\mathfrak{S}$. We omit elements $v_i(j)$ of the permutation in the region if $v_i(j) = j$, unless this element is circled.

The significance of the bullets is that they are the positions in the polynomial from which the variable is being pulled out. The relationship between $v_{i-1}d_i^{-1}$ and v_i is that $v_{i-1}d_i^{-1} \overset{\mathfrak{c}^*_i(u)+1}{\searrow} v_i$, so that this is, in a sense, a path of permutations and the bullet occupies the position of $n + 1$.

Example 1.1.13. The permutation $u = [4, 3, 1, 2]$ is dominant, and $\mathfrak{c}(u) = (3, 2)$. Also, $\mathfrak{c}^*(u) = (2, 2, 1)$. Let $v = [3, 1, 4, 5, 2]$ and let $w = [6, 3, 1, 4, 2, 5]$. Then Theorem 1.1 lets us compute the coefficient $c^w_{uv}(y;z)$. We have that $\mathfrak{c}^*(w) = (2, 3, 1, 1, 1)$.

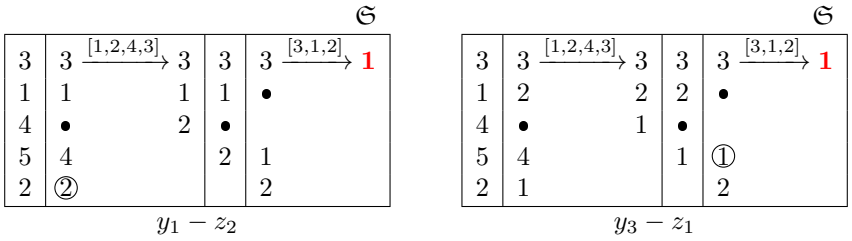


FIGURE 1. Diagrams for first coefficient in Example 1.1.13

Thus $c_{uv}^w(y;z) = (y_1 - z_2) + (y_3 - z_1)$. If we take $w = [6,3,1,2,4,5]$ instead, then $\mathfrak{c}^*(w) = (2,2,1,1,1)$. Thus, we have the following diagrams.

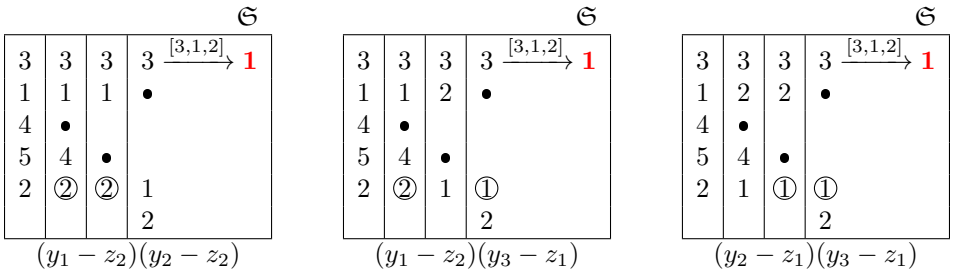


FIGURE 2. Diagrams for second coefficient in Example 1.1.13

Thus $c_{uv}^w(y;z) = (y_1 - z_2)(y_2 - z_2) + (y_1 - z_2)(y_3 - z_1) + (y_2 - z_1)(y_3 - z_1)$.

Example 1.1.14. Let $u = [4,1,2,3]$ and let $v = [1,3,5,2,4]$. We compute $c_{uv}^w(y;z)$ where $w = [4,1,5,2,3]$. We have

$$\mathfrak{c}^*(w) = (1,2,2)$$

$$\mathfrak{c}^*(u) = (1,1,1)$$

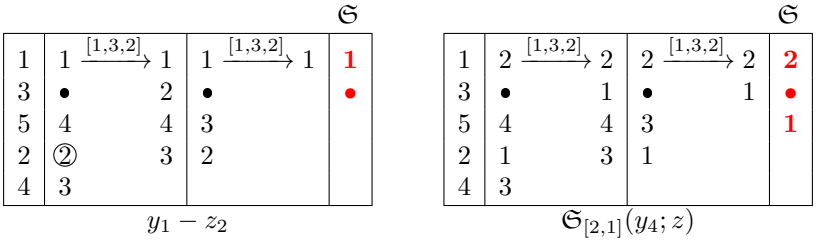


FIGURE 3. Diagrams for Example 1.1.14

Thus, the result is

$$c_{uv}^w(y;z) = (y_1 - z_2) + (y_4 - z_1)$$

Example 1.1.15. We compute $c_{[4,2,1,3],[1,3,5,2,4]}^{[5,2,3,1,4]}(y;z)$. We have

$$\mathfrak{c}^*(w) = (3,1,1,1)$$

$$\mathfrak{c}^*(u) = (2,1,1)$$

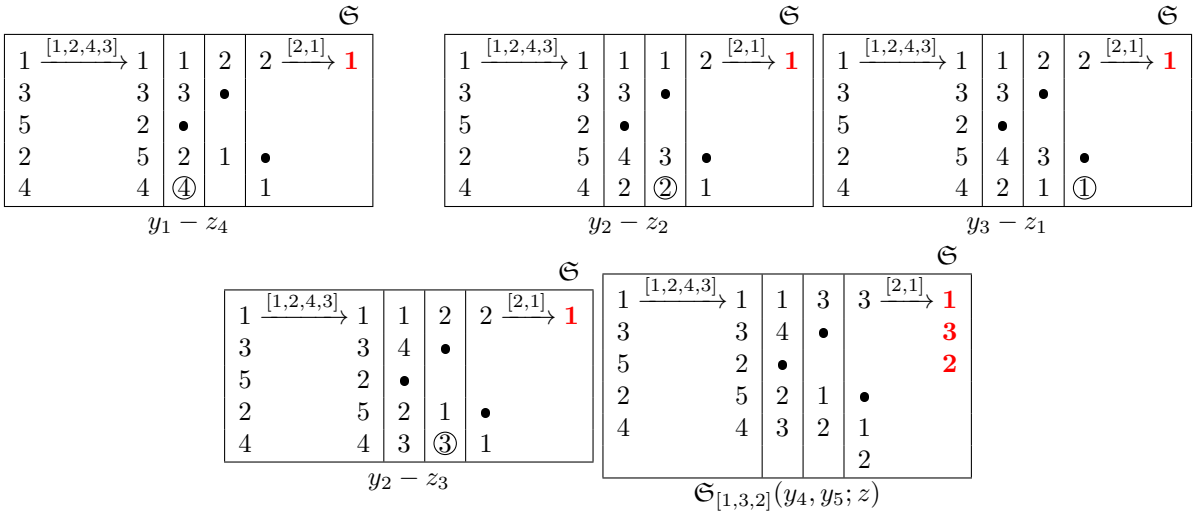


FIGURE 4. Diagrams for Example 1.1.15

Thus, the result is $c_{uv}^{w}(y;z) = y_1 + 2y_2 + y_3 + y_4 + y_5 - 2z_1 - 2z_2 - z_3 - z_4$.

Example 1.1.16. We give an example where u is not dominant. Set $u = [2,3,5,1,4]$, set $v = [2,5,3,4,7,1,6]$, and set $w = [2,6,3,4,7,1,5]$. Then

$$\mathfrak{c}^*(w) = (5,0,1,1,2)$$

$$\mathfrak{c}^*(u) = (3,0,0,1)$$

Then u dominates w but is not dominant. We have the following diagrams.

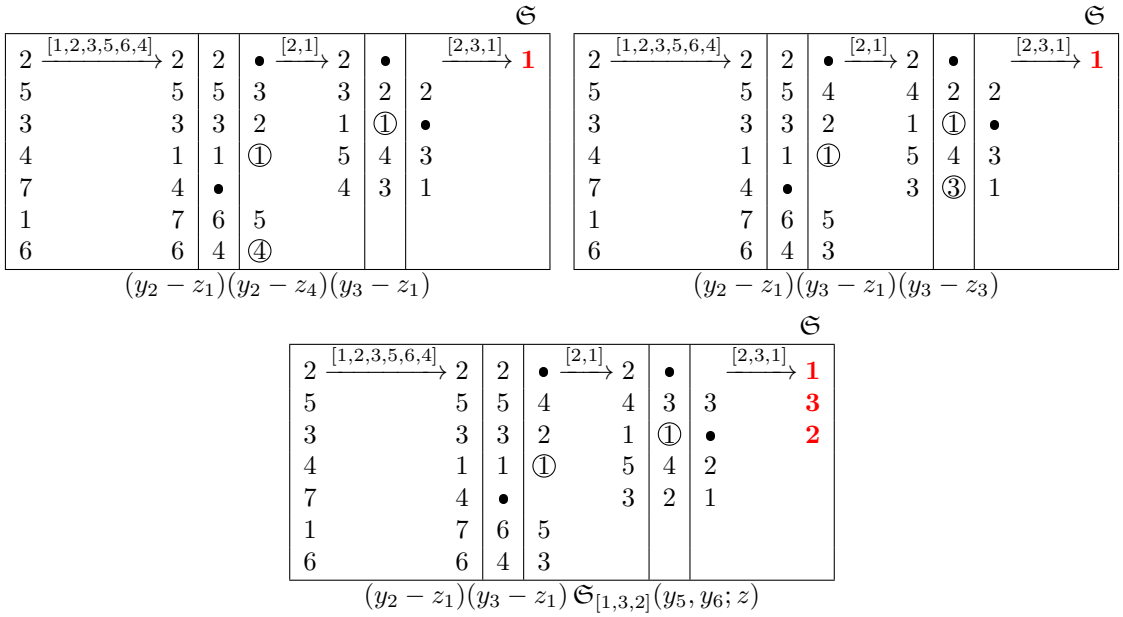


FIGURE 5. Diagrams for Example 1.1.16

Thus

$$c_{uv}^{w}(y;z) = (y_2 - z_1)(y_2 - z_4)(y_3 - z_1) + (y_2 - z_1)(y_3 - z_1)(y_3 - z_3) + (y_2 - z_1)(y_3 - z_1)(y_5 + y_6 - z_1 - z_2)$$

Example 1.1.17. We now compute $c_{[5,4,3,2,1],[4,1,3,5,2]}^{[7,5,3,2,1,4,6]}(y;z)$. We have

$$\mathfrak{c}^*(w) = (4,3,2,2,1,1)$$

$$\mathfrak{c}^*(u) = (4,3,2,1)$$

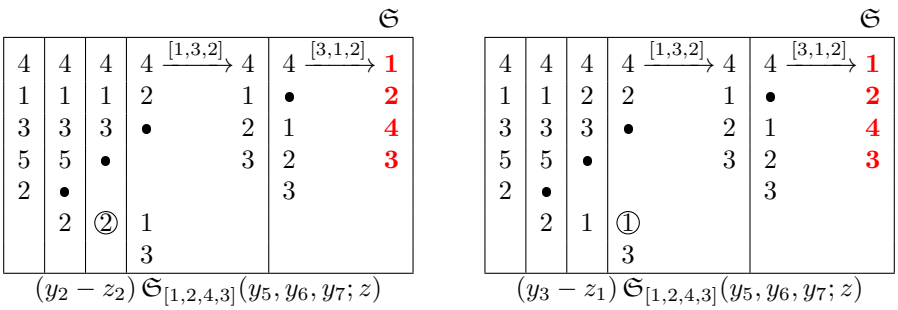


FIGURE 6. Diagrams for Example 1.1.17

Thus

$$c^w_{u,v}(y;z) = (y_2 - z_2)(y_5 + y_6 + y_7 - z_1 - z_2 - z_3) + (y_3 - z_1)(y_5 + y_6 + y_7 - z_1 - z_2 - z_3)$$

1.2. **Dominant permutations, the code, and manipulation of variables in Schubert polynomials.** Here we gather some results about dominant permutations, the code, the dual code, and the function ϕ_i that will be useful throughout this article.

Definition 1.2.1 (ρ_m)**.** For each $m \geq 0$, there are homomorphisms $\rho_m : S_\infty \to S_\infty$ such that

$$\rho_m(s_i) = \begin{cases} s_i & \text{if } i < m \\ t_{m,m+2} & \text{if } i = m \\ s_{i+1} & \text{otherwise.} \end{cases}$$

Essentially, $\rho_m(u)$ is the permutation u' such that

$$u'(i) = \begin{cases} u(i) + 1 & \text{if } i \leq m \\ u(i-1) + 1 & \text{if } i > m+1 \\ 1 & \text{if } i = m+1 \end{cases}$$

Lemma 1.2.2. *For $u \in S_\infty$ we have*

$$\phi(u) = \rho^{-1}_{\mathfrak{c}^*_1(u)}(u)$$

Proof. $\mathfrak{c}^*_1(u)$ is equal to the number of inversions (i,j) of u such that $u(j) = 1$. By the characterization of $\mathfrak{c}^*_1$ near the end of Definition 1.1.5, which is to say that $u(\mathfrak{c}^*_1(u)+1) = 1$, it follows that $\mathfrak{c}^*(\rho_{\mathfrak{c}^*_1(u)}(\phi(u))) = \mathfrak{c}^*(u)$. Thus $\rho_{\mathfrak{c}^*_1(u)}(\phi(u)) = u$, and the result follows. □

The following follows easily.

Corollary 1.2.3. *Let $u \in S_\infty$ and let $r = \mathfrak{c}^*_1(u) + 1$. If $j < r$, then*

$$\mathfrak{c}_j(\phi(u)) = \mathfrak{c}_j(u) - 1$$

If $j \geq r$, then

$$\mathfrak{c}_j(\phi(u)) = \mathfrak{c}_{j+1}(u)$$

Lemma 1.2.4. *A permutation $u \in S_\infty$ is dominant if and only if for all permutations $w \in S_\infty$ we have that u dominates w.*

Proof. First, we prove that if $u \in S_\infty$ is dominant, then for any $w \in S_\infty$, we have that u dominates w. Let i be the index such that $u(i) = 1$, which is $\mathfrak{c}^*_1(u) + 1$ by Lemma 1.2.2. Then $\mathfrak{c}_i(u) = 0$. Since u is dominant, we have that $\mathfrak{c}_j(u) = 0$ for all $j \geq i$ as well, so that for all $j \geq i$ we have that $u(j) < u(j+1)$. This means that for any permutation $w \in S_\infty$, we have that u 1-dominates w. Now, for any $m \geq 0$ we have that $\phi_m(u)$ is dominant. Thus $\phi_m(u)$ 1-dominates $\phi_m(w)$ for all $m \geq 0$, regardless of what w is, so that u dominates w. Thus, if u is dominant, then u dominates all permutations w.

Now suppose $u \in S_\infty$ dominates all permutations. Then $u(j) < u(j+1)$ for all $j > \mathfrak{c}^*_1(u)$. This means that every index j after the first index i such that $\mathfrak{c}_i(u) = 0$ (by Lemma 1.2.2) satisfies $\mathfrak{c}_j(u) = 0$. By definition,

$\phi(u)$ dominates all permutations as well. By the induction hypothesis, $\mathfrak{c}(\phi(u))$ is a partition, and we have that

$$\mathfrak{c}_j(u) = \begin{cases} \mathfrak{c}_j(\phi(u)) + 1 & \text{if } j \leq \mathfrak{c}_1^*(u) \\ 0 & \text{if } j > \mathfrak{c}_1^*(u) \end{cases}$$

Thus $\mathfrak{c}(u)$ is a partition, and the result follows. □

It will be useful to have a formula for $\mathfrak{c}^*(\mu)$ in terms of $\mathfrak{c}(\mu)$ for μ dominant.

Lemma 1.2.5. *Suppose μ is a dominant permutation. Then μ^{-1} is dominant and $\mathfrak{c}^*(\mu)$ is the conjugate partition of $\mathfrak{c}(\mu)$, that is to say*

$$\mathfrak{c}_i^*(\mu) = \#\{j \mid \mathfrak{c}_j(\mu) \geq i\}$$

Proof. By the discussion in Definition 1.1.5, we have that

$$\mathfrak{c}_1^*(\mu) = \#\{j \mid \mathfrak{c}_j(\mu) \geq 1\}$$

We prove the result by induction on $\ell(\mu)$, the base case of $\ell(\mu) = 0$ being clear. We have that

$$\mathfrak{c}_i^*(\phi(\mu)) = \mathfrak{c}_{i+1}^*(\mu)$$

for all i, and by Corollary 1.2.3 we have that

$$\mathfrak{c}_i(\phi(\mu)) = \mathfrak{c}_i(\mu) - 1$$

for all i such that $\mathfrak{c}_i(\mu) > 0$. By the induction hypothesis, for each $i \geq 1$ we have

$$\mathfrak{c}_i^*(\phi(\mu)) = \#\{j \mid \mathfrak{c}_j(\mu) - 1 \geq i\} = \#\{j \mid \mathfrak{c}_j(\mu) \geq i + 1\}$$

Thus for $i > 1$ we have

$$\mathfrak{c}_i^*(\mu) = \mathfrak{c}_{i-1}^*(\phi(\mu)) = \#\{j \mid \mathfrak{c}_j(\mu) \geq i\}$$

The result follows by induction. □

Definition 1.2.6 ($*$, ∇^v)**.** We wish now to apply divided difference operators to the y variables. We thus define an action $*$ of S_∞ on the polynomial ring permuting the indices of the y variables, and define

$$\nabla^{s_i}(P) = \frac{1}{y_i - y_{i+1}}(s_i - 1) * P$$

Note that we have flipped the sign of the divided difference operator to have the formulas

$$\nabla^{s_i}(\mathfrak{S}_v(x; y)) = 0$$

if $\ell(s_i v) > \ell(v)$, and

$$\nabla^{s_i}(\mathfrak{S}_v(x; y)) = \mathfrak{S}_{s_i v}(x; y)$$

if $\ell(s_i v) < \ell(v)$. We define skew divided difference operators ∇_u^w similarly to

$$\nabla^w(PQ) = \sum_u \nabla^u(P)\nabla_u^w(Q)$$

The basis of our formula is the Pieri formula given in [22], of which we quote a special case.

Proposition 1.2.7 (Special case of Pieri formula [22, Theorem 7.1])**.** *Suppose $k \geq 1$ and $u, w \in S_\infty$. If $u \overset{k}{\not\to} w$, then*

$$\nabla_u^w \prod \left(x_1 \mid y_{[k]}\right) = 0$$

If $u \overset{k}{\to} w$, define

$$Q = \{u(i) \mid i \leq k \text{ and } u(i) = w(i)\}$$

then

$$\nabla_u^w \prod \left(x_1 \mid y_{[k]}\right) = \prod \left(x_1 \mid y_Q\right)$$

Proof. This is simply a change of variables from the original theorem. □

The next proposition specialize, in the case of ordinary Schubert polynomial $\mathfrak{S}_v(x)$, to a formular that isolate a chosen index i: one can express $\mathfrak{S}_v(x)$ as a sum of terms of the form $x_i^p\, \mathfrak{S}_{v'}(x^{(i)})$, thereby effectively extracting the variable x_i and leaving Schubert polynomials in the remaining variables [3, Theorem 5.1]. Proposition 1.2.8 is the double Schubert polynomial version of this, which, as far as we know, is new.

Proposition 1.2.8. *Let $v \in S_\infty$ and let $i > 0$ be an integer. Then we have*

$$\mathfrak{S}_v(x;y) = \sum_{v \overset{i}{\searrow} v'} \prod \left(x_i \mid y_{Q_i(v',v)}\right) \mathfrak{S}_{v'}(x^{(i)};y)$$

Proof. Suppose $v \in S_n$. We have

$$\mathfrak{S}_v(x;y) = \nabla^{vw_0(n)}(\mathfrak{S}_{w_0(n)}(x;y))$$

This is equal to

$$\nabla^{vw_0(n)}(\mathfrak{S}_{s_{n+1-i}\cdots s_1 w_0(n)}(x^{(i)};y)\prod \left(x_i \mid y_{[n+1-i]}\right))$$

and, applying the Leibniz formula, is also equal to

$$\sum_{\substack{v' \in S_\infty \\ \ell(v'w_0(n)s_1\cdots s_{n+1-i}) = \ell(w_0(n)s_1\cdots s_{n+1-i}) - \ell(v')}} \mathfrak{S}_{v'}(x^{(i)};y)\nabla^{vw_0(n)}_{v'w_0(n)s_1\cdots s_{n+1-i}} \prod \left(x_i \mid y_{[n+1-i]}\right)$$

By the restricted Pieri formula (Proposition 1.2.7), for this to be nonzero necessarily $v'w_0(n)s_1\cdots s_{n+1-i} \xrightarrow{n+1-i} vw_0(n)$. We note that $v'w_0(n)s_1\cdots s_{n+1-i} \xrightarrow{n+1-i} vw_0(n)$ if and only if

$$v \overset{i}{\Rightarrow} v'w_0(n)s_1\cdots s_{n+1-i}w_0(n) = v's_ns_{n-1}\cdots s_i$$

We have that $v's_n\cdots s_i$ is exactly $\varphi_{i,n}(v')$ since $v' \in S_n$, so we require that

$$v \overset{i}{\Rightarrow} \varphi_{i,n}(v')$$

so the sum is over all $v' \in \mathcal{D}_i(v)$.

Applying Proposition 1.2.7, we obtain that the result is equal to

$$\sum_{v' \in \mathcal{D}_i(v)} \prod \left(x_i \mid y_{A(v',v)}\right) \mathfrak{S}_{v'}(x^{(i)};y)$$

where $A(v',v)$ is the set of all $vw_0(n)(j)$ such that $1 \le j \le n+1-i$ and $v'w_0(n)s_1\cdots s_{n+1-i}(j) = vw_0(n)(j)$. These values are the same as at the indices that comprise the set of all $1 \le j \le n+1-i$ such that

$$v'(n+2-s_1\cdots s_{n+1-i}(j)) = v(n+2-j)$$

Applying the $s_1\cdots s_{n+1-i}$ to j, since $1 \le j \le n+1-i$ we have that

$$s_1\cdots s_{n+1-i}(j) = j+1$$

Hence we need

$$v'(n+2-(j+1)) = v'(n+1-j) = v(n+2-j)$$

Replacing j with $n+2-p$, the indices are the set of all p such that $i < p \le n+1$ and

$$v'(p-1) = v(p)$$

Thus $A(v',v)$ is the set of all $v(p)$ such that $p > i$ and $v'(p-1) = v(p)$, which is exactly $Q_i(v',v)$, and we are done. □

Example 1.2.9. Suppose $i = 2$ and $v = [2,5,1,3,4]$. Then

$$\mathcal{D}_i(v) = \{[2,1],[3,1,2],[4,1,2,3]\}$$

$$Q_i([2,1],[2,5,1,3,4]) = \{1,3,4\}$$

$$Q_i([3,1,2],[2,5,1,3,4]) = \{1,4\}$$

$$Q_i([4,1,2,3],[2,5,1,3,4]) = \{1\}$$

Thus

$$\mathfrak{S}_{[2,5,1,3,4]}(x;y) = \prod (x_2 \mid y_1,y_3,y_4)\,\mathfrak{S}_{[2,1]}(x^{(2)};y) + \prod (x_2 \mid y_1,y_4)\,\mathfrak{S}_{[3,1,2]}(x^{(2)};y) + \prod (x_2 \mid y_1)\,\mathfrak{S}_{[4,1,2,3]}(x^{(2)};y)$$

Example 1.2.10. Suppose $i = 3$ and $v = [2, 4, 6, 5, 3, 1]$. Then

$$\mathcal{D}_i(v) = \{[2, 4, 5, 3, 1], [3, 4, 5, 2, 1], [2, 5, 4, 3, 1], [3, 5, 4, 2, 1]\}$$
$$Q_i([2, 4, 5, 3, 1], [2, 4, 6, 5, 3, 1]) = \{1, 3, 5\}$$
$$Q_i([3, 4, 5, 2, 1], [2, 4, 6, 5, 3, 1]) = \{1, 5\}$$
$$Q_i([2, 5, 4, 3, 1], [2, 4, 6, 5, 3, 1]) = \{1, 3\}$$
$$Q_i([3, 5, 4, 2, 1], [2, 4, 6, 5, 3, 1]) = \{1\}$$

Thus

$$\begin{aligned}\mathfrak{S}_{[2,4,6,5,3,1]}(x; y) =& \prod(x_3 \mid y_1, y_3, y_5)\, \mathfrak{S}_{[2,4,5,3,1]}(x^{(3)}; y) + \prod(x_3 \mid y_1, y_5)\, \mathfrak{S}_{[3,4,5,2,1]}(x^{(3)}; y) \\ &+ \prod(x_3 \mid y_1, y_3)\, \mathfrak{S}_{[2,5,4,3,1]}(x^{(3)}; y) + \prod(x_3 \mid y_1)\, \mathfrak{S}_{[3,5(x_3;y_1),4,2,1]}(x^{(3)}; y)\end{aligned}$$

Example 1.2.11. We compute $\mathfrak{S}_{[1,4,3,2]}(x; y)$ with Theorem 1.2.

1	•		
4	3	•	
3	2	②	•
2	1	①	①

$(x_2 - y_1)(x_2 - y_2)(x_3 - y_1)$

1	•		
4	2	•	
3	③	2	•
2	1	①	①

$(x_1 - y_3)(x_2 - y_1)(x_3 - y_1)$

1	•		
4	3	•	
3	1	①	
2	②	②	

$(x_1 - y_2)(x_2 - y_1)(x_2 - y_2)$

1	•		
4	1	•	
3	③	2	•
2	②	1	①

$(x_1 - y_2)(x_1 - y_3)(x_3 - y_1)$

1	•		
4	1	•	
3	③	1	
2	②	②	

$(x_1 - y_2)(x_1 - y_3)(x_2 - y_2)$

FIGURE 7. Diagrams for computation of double Schubert polynomial in Example 1.2.11

1.3. **Relation to pipe dreams.** Proposition 1.2.8 leads to the following positive formula for Schubert polynomials.

Theorem 1.2. *Let $v \in S_\infty$ with $\ell(vs_i) > \ell(v)$ for all $v > n$ for some $n > 0$. Then*

$$\mathfrak{S}_v(x; y) = \sum_{(v_j)_{j=0}^n} \prod_{i=1}^n \prod(x_i \mid y_{Q_1(v_i, v_{i-1})})$$

where the sequence of v_i ranges over all such that $v_0 = v$ and $v_{i-1} \overset{1}{\searrow} v_i$ for all $i \geq 1$.

The significance of Theorem 1.2 lies in its ability to connect the monomials of double Schubert polynomials (products of terms of the form $x_i - y_j$) to reduced words with compatible sequences when expressed in this form.

Lemma 1.3.1. *Let v be a permutation. If $v \overset{1}{\searrow} v'$, let $a_1, \ldots, a_k$ be the elements of $Q_1(v', v)$ in decreasing order. Then*

$$v = s_{a_1} \cdots s_{a_k} \sigma(v')$$

Proof. Suppose $v \in S_n$. We have by definition that there is a sequence of integers $b_1, \ldots, b_p$, all distinct and greater than 1, such that

$$v t_{1,b_1} \cdots t_{1,b_p}(1) = n + 1$$

and

$$v t_{1,b_1} \cdots t_{1,b_p}(i + 1) = v'(i)$$

for all $i < n - 1$. This means that

$$v t_{1,b_1} \cdots t_{1,b_p} = s_n s_{n-1} \cdots s_1 \sigma(v')$$

This is because if $v'' = \sigma(v')$, then

$$v''(1) = 1$$

and

$$v''(i) = v(i - 1) + 1$$

The cycle $s_n \cdots s_1$ sends $1 \mapsto n+1$ and $i \mapsto i-1$ if $1 < i \leq n+1$, hence

$$vt_{1,b_1} \cdots t_{1,b_p} = s_n s_{n-1} \cdots s_1 \sigma(v')$$

In particular,

$$v = s_n s_{n-1} \cdots s_1 \sigma(v') t_{1,b_p} \cdots t_{1,b_1}$$

This results in the factorization

$$v = s_n \cdots s_1 t_{1,v'(b_p-1)+1} \cdots t_{1,v'(b_1-1)+1} \sigma(v')$$

The value $v'(b_j - 1)$ necessarily decreases as j decreases, since applying the corresponding reflection in the reverse order strictly increases the length with each application. Consequently, multiplying $s_n \cdots s_1$ on the right by these reflections remove the simple reflections $s_{v'(b_p-1)}, \ldots, s_{v'(b_1-1)}$. The indices removed are precisely the complement of the elements of $Q_1(v', v)$, hence the elements of $Q_1(v', v)$ in decreasing order are what remain, as desired. □

Theorem 1.3. *Suppose $v = v_0 \overset{1}{\searrow} v_1 \cdots \overset{1}{\searrow} v_{k+1} = 1$. Let $q_{i,1}, \ldots q_{i,m_i}$ be the elements of $Q_1(v_i, v_{i-1})$ in decreasing order. Let the sequences $a_1, \ldots, a_m$ and $b_1, \ldots, b_m$ be the concatenation, in order of increasing i, of the sequences of length m_i*

$$q_{i,1} + i - 1, \ldots, q_{i,m_i} + i - 1$$

and

$$(i, i, \ldots, i)$$

respectively. Then $(a_1, \ldots, a_m)$ are the indices of the simple reflections in a reduced word for v, and $(b_1, \ldots, b_m)$ is a sequence of integers compatible with this reduced word.

1.4. **Proof of the domination formula.** The idea of the proof is to pick apart the divided difference to apply it to the double Schubert polynomial variable index by variable index.

Lemma 1.4.1. *Let $u, w \in S_\infty$, let $p \in \mathbb{Z}[x]$, and let $i > 0$ be an integer. Then*

$$\partial_{\sigma_i(u)}^{\sigma_i(w)}(\tau^i(p)) = \tau^i(\partial_u^w(p))$$

Proof. Since

$$\partial^{s_j} = \frac{1}{x_j - x_{j+1}}(1 - s_j)$$

It suffices to show that

$$\sigma_i(s_j)(\tau^i(x_p)) = \tau^i(s_j(x_p))$$

Writing this out, we have on the left-hand side

$$s_{i+j}(x_{i+p}) = \begin{cases} x_{i+p+1} & \text{if } p = j \\ x_{i+p-1} & \text{if } p = j+1 \\ x_{i+p} & \text{otherwise} \end{cases}$$

which agrees with the right-hand side when worked out. □

Definition 1.4.2. Let $(a_1, \ldots, a_m)$ be a reduced word. A sequence $(b_1, \ldots, b_m)$ of the same length such that $b_i \in \{1, a_i\}$ is said to be a *subword* for the element $b_1 \cdots b_m$. Let $(b_{i_1}, \ldots, b_{i_k})$ be all of the nonunit elements of this subword, with $i_1 < \cdots < i_k$. Then the subword is said to be reduced if $(b_{i_1}, \ldots, b_{i_k})$ is a reduced word.

The following was proved by Macdonald in [17].

Proposition 1.4.3. *Suppose $u, w \in S_\infty$ and let $(a_1, \ldots, a_m)$ be a fixed reduced word for w. Then ∂_u^w is the sum over all reduced subwords $(b_1, \ldots, b_m)$ for u of the elements*

$$\partial_{b_1}^{a_1} \partial_{b_2}^{a_2} \cdots \partial_{b_m}^{a_m}$$

Proof. This is true if $w = 1$. Otherwise, suppose $(a_1, \ldots, a_m)$ is a reduced word for w. Let

$$\eta(u, a_m) = \begin{cases} 1 & \text{if } \ell(ua_m) < \ell(u) \\ 0 & \text{if } \ell(ua_m) > \ell(u) \end{cases}$$

Then

$$\partial_u^w = \partial_u^{wa_m} \partial^{a_m} + \eta(u, a_m) \partial_{ua_m}^{wa_m} a_m$$

The set of reduced subwords $(b_1, \ldots, b_m)$ for u is the union of the set of all subwords $(b_1, \ldots, b_{m-1}, 1)$ that are reduced subwords for u plus the set of all subwords $(b_1, \ldots, b_{m-1}, a_m)$ such that $(b_1, \ldots, b_{m-1})$ is a reduced subword of $(a_1, \ldots, a_{m-1})$ for ua_m, with this second set occurring only if $\eta(u, a_m) = 1$. Let

$$D = \partial_{b_1}^{a_1} \cdots \partial_{b_{m-1}}^{a_{m-1}}$$

D occurs in $\partial_u^{wa_m}$ where $(b_1, \ldots, b_{m-1})$ is a reduced subword for u by the induction hypothesis, and $\partial^{a_m} = \partial_1^{a_m}$, so the terms in the summand $\partial_u^{wa_m} \partial^{a_m}$ correspond bijectively to subwords of the first type. For the second summand, D occurs in $\partial_{ua_m}^{wa_m}$ where $(b_1, \ldots, b_{m-1})$ is a reduced subword for ua_m by the induction hypothesis, and $a_m = \partial_{a_m}^{a_m}$, so the terms in the summand $\partial_{ua_m}^{wa_m} a_m$ correspond bijectively to subwords of the second type. Since this accounts for all of them, the result follows by induction. □

Lemma 1.4.4. *Let $w \in S_\infty$ have $\mathfrak{c}^*(w) = (m_1, \ldots, m_q)$. Then*

$$w = d[q, m_q] \cdots d[2, m_2] d[m_1]$$

Proof. This is true if $\ell(w) = 0$. Otherwise, suppose $\ell(w) > 0$ and the result holds for smaller lengths. Assume without loss of generality that $m_q > 0$. Then

$$s_q w = d[q+1, m_q - 1] d[q-1, m_{q-1}] \cdots d[2, m_2] d[m_1]$$

by the induction hypothesis, since

$$\mathfrak{c}^*(s_q w) = (m_1, \ldots, m_{q-1}, 0, m_q - 1)$$

Since $s_q d[q+1, m_q - 1] = d[q, m_q]$, the result follows for $s_q(s_q w) = w$ by induction. □

Definition 1.4.5 (The normal form)**.** The element $d[a, b]$ (for all $a \geq 1$, $b \geq 0$) has a unique reduced word. Consequently, the factorization $w = d[q, m_q] \cdots d[m_1]$ corresponds to an unambiguous reduced word for w. We call this reduced word the *normal form* of w. This is simply a convenient choice of normal form for our situation. More generally, systematic procedures exist for computing normal forms of elements in Coxeter groups via parabolic subgroups; the construction above is one instance of such methods in the case of symmetric groups [4].

Lemma 1.4.6. *Let $(a_1, \ldots, a_m)$ be the normal form for an element $w \in S_\infty$ and let $(b_1, \ldots, b_m)$ be a reduced subword for an element $u \in S_\infty$ with $\mathfrak{c}^*(u) = (m_1', \ldots, m_p')$. Then $m_1' \leq m_1$. If $(a_j, a_{j+1}, \ldots, a_m)$ is the ending portion such that $a_j \cdots a_m = d[m_1]$, then if $m_1' > 0$ we have $b_i \neq 1$ for all $j \leq i < j + m_1'$.*

Proof. Since $d[m_1]$ and $d[m_1']$ are the minimal length left coset representatives of w and u, respectively, with respect to the parabolic subgroup generated by $\{s_i\}_{i>1}$, it follows that $d[m_1'] \leq d[m_1]$ in Bruhat order. Hence $m_1' \leq m_1$ (see [4]). We prove the subword statement by induction on $\ell(u)$. **Base cases** If $m_1' = 0$, there is nothing to prove. If 1 is the only index i such that $m_i' \neq 0$, then we can see that this is true for combinatorial reasons ($d[m_1']$ has exactly one reduced word, and the only place it fits is at the beginning of $(a_j, \ldots, a_m)$). **Induction Step** Suppose $p > 1$ is the last index such that $m_p' > 0$. Then $\ell(s_p u) < \ell(u)$, so by setting one nonunit element to 1 in $(b_1, \ldots, b_m)$ we obtain a reduced subword for $s_p u$ (by the exchange property). By the induction hypothesis, the desired portion of $(a_j, \ldots, a_m)$ is occupied by m_1' nonunit elements in this reduced subword for $s_p u$ since $\mathfrak{c}_1^*(s_p u) = m_1'$. Replacing the element removed by multiplying on the left by s_p, we obtain the result. □

Lemma 1.4.7. *Let $u, w \in S_\infty$ and suppose u 1-dominates w. Then $\partial_u^w = 0$ unless $\mathfrak{c}_1^*(w) \geq \mathfrak{c}_1^*(u)$, in which case*

$$\partial_u^w = \partial_{\sigma(\phi(u))}^{\sigma(\phi(w))} d[\mathfrak{c}_1^*(u)] \partial^{d[\mathfrak{c}_1^*(u)+1, \mathfrak{c}_1^*(w) - \mathfrak{c}_1^*(u)]}$$

Proof. Let $(a_j, \ldots, a_m)$ be the consecutive ending portion of the normal form of w such that $a_j = s_1$ and let $(b_1, \ldots, b_m)$ be a reduced subword for u. By Lemma **??**, $b_p \neq 1$ for all $j \leq p < j + \mathfrak{c}_1^*(u)$ (whereas if $\mathfrak{c}_1^*(w) < \mathfrak{c}_1^*(u)$, this is not possible and there is no reduced subword for u). Since u has no descent between $\mathfrak{c}_1^*(u)$ and $\mathfrak{c}_1^*(w)$, we have that $b_p = 1$ for all $p \geq j + \mathfrak{c}_1^*(u)$. It follows that

$$\partial_u^w = \partial_{ud[\mathfrak{c}_1^*(u)]^{-1}}^{wd[\mathfrak{c}_1^*(w)]^{-1}} d[\mathfrak{c}_1^*(u)] \partial^{d[\mathfrak{c}_1^*(u)+1, \mathfrak{c}_1^*(w) - \mathfrak{c}_1^*(u)]}$$

since this is true for every subword. Now, we can see by examining the normal forms of these elements that

$$ud[\mathfrak{c}_1^*(u)]^{-1} = \sigma(\phi(u))$$

and

$$wd[\mathfrak{c}_1^*(w)]^{-1} = \sigma(\phi(u))$$

and the result follows. □

Proof of Theorem 1.1. We compute this by induction on the maximal index m such that $\mathfrak{c}_m^*(u) \neq 0$, with the base case being that u is the identity. If u is the identity, then the result is

$$\partial^w(\mathfrak{S}_v(x; z))|_{x=y}$$

This agrees with the stated formula when we set $m = 0$. For the induction step, by Lemma 1.4.7 we have

$$\partial_u^w = \partial_{\sigma(\phi(u))}^{\sigma(\phi(w))} d[\mathfrak{c}_1^*(u)] \partial^{d_1}$$

We apply this to the double Schubert polynomial $\mathfrak{S}_v(x; z)$. Applying ∂^{d_1}, we obtain $\mathfrak{S}_{vd_1^{-1}}(x; z)$, provided $\ell(vd_1^{-1}) = \ell(v) - \ell(d_1)$, or otherwise 0. Assuming the term is nonzero, using Proposition 1.2.8, we have that

$$\mathfrak{S}_{vd_1^{-1}}(x; z) = \sum_{v_1 \in \mathcal{D}_{\mathfrak{c}_1^*(u)+1}(vd_1^{-1})} \prod \left(x_{\mathfrak{c}_1^*(u)+1} \mid z_{A_1}\right) \mathfrak{S}_{v_1}(x^{(\mathfrak{c}_1^*(u)+1)}; z)$$

Now applying $d[\mathfrak{c}_1^*(u)]$, this becomes

$$\sum_{v_1 \in \mathcal{D}_{\mathfrak{c}_1^*(u)+1}(vd_1^{-1})} \prod (x_1 \mid z_{A_1}) \mathfrak{S}_{v_1}(\tau x; z)$$

Now we are trying to compute $\partial_{\sigma(\phi(u))}^{\sigma(\phi(w))}$ applied to this sum. By Lemma 1.4.1, the application to the double Schubert polynomial $\mathfrak{S}_{v_1}(\tau x; z) = \mathfrak{S}_{v_1}(\tau x; z)$ is equal to

$$\tau \partial_{\phi(u)}^{\phi(w)}(\mathfrak{S}_{v_1}(x; z))$$

We can apply the induction hypothesis to compute this, since $\phi(u)$ dominates $\phi(w)$ and $\mathfrak{c}^*(\phi(u))$ is strictly shorter than $\mathfrak{c}^*(u)$ (in the sense of the rightmost position of nonzero elements), and theref we have the result. □

1.5. **Reversing the separated descents condition.** The article [22] uses the properties of dominant permutations and the Pieri formula to compute products of the form $\mathfrak{S}_u(x; y) \mathfrak{S}_v(x; z)$ where there exists a p with $\ell(us_i) > \ell(u)$ if $i < p$ and $\ell(vs_i) > \ell(v)$ if $i > p$. We apply Theorem 1.1 to reverse the conditions in this formula.

Lemma 1.5.1. *Let $u, v, w \in S_\infty$ and let s_i be such that none of u, v, w has s_i as a descent. Then*

$$c_{u,v}^w(y; z) = c_{us_i,v}^{ws_i}(y; z)$$

and

$$c_{u,v}^w(y; z) = c_{u,vs_i}^{ws_i}(y; z)$$

Proof. The latter formula was demonstrated in [22]. For the former, we apply $\partial_{us_i}^{ws_i}$ to $\mathfrak{S}_v(x; z)$. We have that

$$\partial_{us_i}^{ws_i} = \partial_{us_i}^w \partial^{s_i} + \partial_u^w s_i$$

We have that $\partial^{s_i}(\mathfrak{S}_v(x; z)) = 0$, and $s_i(\mathfrak{S}_v(x; z)) = \mathfrak{S}_v(x; z)$, so the end result is

$$\partial_u^w(\mathfrak{S}_v(x; z))$$

as desired. □

The following theorem is dual to Theorem 8.1 in [22].

Theorem 1.4. *Let $u, v, w \in S_\infty$. Suppose $u^{-1}\mathfrak{d}(u) = s_{i_1} \cdots s_{i_k}$ with $\ell(vs_{i_j}) > \ell(v)$ for all $1 \leq j \leq k$. Then $c^w_{u,v}(y; z) = 0$ unless $\ell(wu^{-1}\mathfrak{d}(u)) = \ell(w) + \ell(u^{-1}\mathfrak{d}(u))$, in which case*

$$c^w_{u,v}(y; z) = c^{wu^{-1}\mathfrak{d}(u)}_{\mathfrak{d}(u),v}(y; z)$$

Hence, this can be computed with Theorem 1.1 and has nonnegative integer coefficients as a polynomial in the terms $y_i - z_j$.

In particular, this applies whenever there exists a p such that $\ell(u) < \ell(us_i)$ for all $i > p$ and $\ell(v) < \ell(vs_i)$ for all $i < p$.

Proof. The proof is identical to that of [22, Proposition 4.3], hence we cite that proof as proof of this result. □

Hence, citing Theorem 3.1 of [22], if u and v have separated descents, in any direction, then $c^w_{u,v}(y; z)$ has nonnegative integer coefficients as a polynomial in $y_i - z_j$. This formula completes all of the generalizations in [22] that could only be generalized in one direction; [22] is the correct direction in which to compute the equivariant case positively, but "commutativity" (not literally, but rather the ability to switch the lower indices) is not present in the general mixed-variable setting. The generalizations of formulas in [19] (multiplying factorial Schur polynomials with different numbers of x variables) and [15] (multiplying a double Schubert polynomial by a factorial Schur polynomial that has at least as many x variables) are made computable positively in the opposite direction with Theorem 1.4, where the factorial Schur polynomial with more x variables has coefficients in z (whereas only coefficients in y were computed previously). This also provides a third way to compute a product of two factorial Schur polynomials with the same number of x variables (the first being in [18], the second in [22]). The resulting formula, however, is not positive when substituting $y = z$, unlike the formula in [22], which is positive in this sense.

Example 1.5.2. We compute a dual example to Example 8.1 in [22], where $u = [3, 1, 4, 2]$ and $v = [4, 1, 3, 2]$. u is not a dominant permutation, but Theorem 1.4 applies. We choose $w = [4, 1, 5, 2, 3]$. Then we have

$$\lambda(u) = (2, 2)$$
$$\mathfrak{d}(u) = [3, 4, 1, 2]$$
$$wu^{-1}\mathfrak{d}(u) = [4, 5, 1, 2, 3]$$
$$\mathfrak{c}^*(wu^{-1}\mathfrak{d}(u)) = (2, 2, 2)$$

$$\mathfrak{S}$$

4	4	4 $\xrightarrow{[3,1,2]}$	**1**
1	2	2	**4**
3	•		**2**
2	1	•	**3**
	3	1	
		3	

$$\mathfrak{S}_{[1,4,2,3]}(y_3, y_4; z)$$

FIGURE 8. Diagram for Example 1.5.2

Thus

$$c^{[4,1,5,2,3]}_{[3,1,4,2],[4,1,3,2]}(y; z) = (y_3 - z_2)(y_3 - z_3) + (y_3 - z_3)(y_4 - z_1) + (y_4 - z_1)(y_4 - z_2)$$

1.6. **Reduction of product coefficients in the case of 1-domination.** The induction step in the proof of Theorem 1.1 is, in itself, useful in computing other types of coefficients. The result that is of use is as follows.

Proposition 1.6.1. *Suppose $u, w \in S_\infty$ are such that u 1-dominates w and that $v \in S_\infty$. If it is not the case that*

$$\ell(vd[\mathfrak{c}^*_1(u) + 1, \mathfrak{c}^*_1(w) - \mathfrak{c}^*_1(u)]^{-1}) = \ell(v) - (\mathfrak{c}^*_1(w) - \mathfrak{c}^*_1(u)),$$

then $c^w_{u,v}(y;z) = 0$. *Otherwise, let* $p = \mathfrak{c}^*_1(u)$ *and let* $d = d[\mathfrak{c}^*_1(u)+1, \mathfrak{c}^*_1(w) - \mathfrak{c}^*_1(u)]$. *Then*

$$c^w_{u,v}(y;z) = \sum_{vd^{-1} \overset{p+1}{\searrow} v'} \prod \left(y_1 \mid z_{Q_{p+1}(v', vd^{-1})}\right) c^{\phi(w)}_{\phi(u),v'}(\tau y; z)$$

Proof. Applying Lemma 1.4.7, we have

$$\partial^w_u = \partial^{\sigma(\phi(w))}_{\sigma(\phi(u))} s_1 s_2 \cdots s_p \partial^{d[\mathfrak{c}^*_1(u)+1, \mathfrak{c}^*_1(w)-\mathfrak{c}^*_1(u)]}$$

The condition on the length of $vd[\mathfrak{c}^*_1(u)+1, \mathfrak{c}^*_1(w) - \mathfrak{c}^*_1(u)]^{-1}$ then arises because otherwise applying this operator yields 0. Assume the condition holds, then. We have that

$$\mathfrak{S}_{vd^{-1}}(x;z) = \sum_{vd^{-1} \overset{p+1}{\searrow} v'} \prod \left(x_{p+1} \mid z_{Q_{p+1}(v', vd^{-1})}\right) \mathfrak{S}_{v'}(x^{(p+1)}; z)$$

by Proposition 1.2.8. Applying $s_1 \cdots s_p$ yields

$$s_1 \cdots s_p(\mathfrak{S}_{vd^{-1}}(x;z)) = \sum_{vd^{-1} \overset{p+1}{\searrow} v'} \prod \left(x_1 \mid z_{Q_{p+1}(v', vd^{-1})}\right) \mathfrak{S}_{v'}(\tau x; z)$$

$\partial^{\sigma(\phi(w))}_{\sigma(\phi(u))}$ then acts on $\mathfrak{S}_{v'}(\tau x; z)$ as $\partial^{\phi(w)}_{\phi(u)}$ on $\mathfrak{S}_{v'}(x;z)$ by Lemma 1.4.1. The result follows. □

Remark. Proposition 1.6.1 is new even for ordinary Schubert polynomials, setting $y = z = 0$.

Proposition 1.6.1 enables us to reduce certain product coefficients incrementally and, in some cases, compute the coefficient positively. Indeed, Theorem 1.1 illustrates this method for a wide class of coefficients where full positive computation is achieved. However, other formulas in our framework may still allow positive computation of coefficients even when domination does not apply.

Example 1.6.2. Here, it will be natural to represent permutations by their code using the code notation $[\![\cdots]\!]$. For example, consider computing $c^{[\![7,0,1]\!]}_{[\![1,0,3]\!],[\![3,0,2,1]\!]}(y;z)$. Since $\mathfrak{c}^*_1([\![1,0,3]\!]) = 1 = \mathfrak{c}^*_1([\![7,0,1]\!])$, Proposition 1.6.1 applies, reducing the computation to coefficients of the form. $c^{[\![6,1]\!]}_{[\![0,3]\!],v'}(\tau y; z)$. At this stage, however, the prospects are not especially promising, as no special methods are currently available for coefficients of this type.

Computing $\mathcal{D}_2([\![3,0,2,1]\!])$, we obtain

$$\{[\![3,0,1]\!], [\![3,1,1]\!], [\![3,2]\!], [\![3,2,1]\!]\}$$

We can compute, using Lemma 1.5.1 on descent position 2, that

$$c^{[\![6,1]\!]}_{[\![0,3]\!],[\![3,0,1]\!]}(y;z) = c^{[\![6]\!]}_{[\![0,0,2]\!],[\![3,0,1]\!]}(y;z) = 0$$

since this is a separated descents product, so that [22, Theorem 3.1] applies, and similarly

$$c^{[\![6,1]\!]}_{[\![0,3]\!],[\![3,1,1]\!]}(y;z) = c^{[\![6]\!]}_{[\![0,0,2]\!],[\![3,1,1]\!]}(y;z) = 0$$

Both $[\![3,2]\!]$ and $[\![3,2,1]\!]$ are dominant, so these coefficients can be computed with [22, Theorem 8.1]. We have

$$c^{[\![6,1]\!]}_{[\![0,3]\!],[\![3,2]\!]}(y;z) = y_1 + y_2 - z_1 - z_2$$

and

$$c^{[\![6,1]\!]}_{[\![0,3]\!],[\![3,2,1]\!]}(y;z) = (y_1 - z_1)(y_2 - z_1)$$

We have that $Q_2([\![3,0,2,1]\!], [\![3,2,1]\!])$ is empty, and $Q_2([\![3,0,2,1]\!], [\![3,2]\!]) = \{2\}$.

The final result is thus

$$\begin{aligned} c^{[\![7,0,1]\!]}_{[\![1,0,3]\!],[\![3,0,2,1]\!]}(y;z) &= (y_1 - z_2) c^{[\![6,1]\!]}_{[\![0,3]\!],[\![3,2]\!]}(\tau y; z) + c^{[\![6,1]\!]}_{[\![0,3]\!],[\![3,2,1]\!]}(\tau y; z) \\ &= (y_1 - z_2)(y_2 + y_3 - z_1 - z_2) + (y_2 - z_1)(y_3 - z_1) \end{aligned}$$

The following result, announced in [**?**], provides a partial solution to the Schubert-Schur problem. It is also an easy consequence of Proposition 1.6.1, and it asserts that Schubert-Schur products are Schubert positive whenever the Schur polynomial is "sufficiently fat."

Theorem 1.5. *Let $v \in S_\infty$, let $k > 0$ be an integer, and let λ be a partition with at most k parts. Let N be the maximum value of 0 and all $\mathfrak{c}_i(v) + i - 1$ for all $i > k$ such that $\mathfrak{c}_i(v) \neq 0$. If $\lambda_i \geq N - k$ for all i with $1 \leq i \leq k$, then the coefficient of $\mathfrak{S}_w(x; y)$ in the product*

$$s_\lambda(x_1, \ldots, x_k; y)\, \mathfrak{S}_v(x; z)$$

admits a positive combinatorial formula in the differences $y_i - z_j$ for all $w \in S_\infty$.

Remark. Theorem 1.5 is new even for ordinary Schubert-Schur products.

Proof. Let u be the permutation such that $\mathfrak{S}_u(x; y) = s_\lambda(x_1, \ldots, x_k; y)$. In particular, $\mathfrak{c}_i(u) = \lambda_{k+1-i}$ if $1 \leq i \leq k$, and $\mathfrak{c}_i(u) = 0$ if $i > k$. Suppose the polynomial $\mathfrak{S}_w(x; y)$ occurs with a nonzero coefficient in this product. By Lemma 1.5.1 applied in reverse, we may assume w is k-Grassmannian and $\mathfrak{S}_v(x; z)$ has at most N x-variables. We prove this by induction on $N - k$, the result being true in the base case $N - k \leq 0$ by [22, Theorem 3.1], since in that case u and v have separated descents.

By virtue of the fact that $c_{u,v}^w(y; z)$ is nonzero, we have that $\mathfrak{c}_i(w) \geq \mathfrak{c}_i(u)$ for all $1 \leq i \leq k$. In particular, $\mathfrak{c}_i^*(u) = \mathfrak{c}_i^*(w) = k$ for all i with $1 \leq i \leq N - k$. Applying Proposition 1.6.1, we have that

$$c_{u,v}^w(y; z) = \sum_{v' \in \mathcal{D}_{k+1}(v)} \prod \left(y_1 \mid z_{Q_{k+1}(v',v)}\right) c_{\phi(u),v'}^{\phi(w)}(\tau y; z)$$

We have that $\phi(u)$ is k-Grassmannian with partition λ' such that $\lambda'_i = \lambda_i - 1$ for all i, and $\mathfrak{S}_{v'}(x; z)$ has at most $N - 1$ x-variables. We have that

$$\lambda_i - 1 \geq N - 1 - k$$

for all i with $1 \leq i \leq k$. Hence, the induction hypothesis applies to $c_{\phi(u),v'}^{\phi(w)}(y; z)$ for each v' occurring in this sum. It follows that all of the terms in the sum are positive; hence, the result follows by induction. □

2. Product–splitting equivalence

2.1. **The equivalence theorem.** Proposition 1.2.8 describes how a double Schubert polynomial expands after splitting its variables. We now express this expansion in terms of double Schubert polynomials in the two sets of variables.

Definition 2.1.1 (Splitting coefficients)**.** Let A be a set of positive integers and let B be its complement. We define the $\mathbb{Z}[y, z]$-linear variable-splitting map

$$\mathrm{sp}_A : \mathbb{Z}[x, y, z] \to \mathbb{Z}[x, y, z] \otimes_{\mathbb{Z}[y,z]} \mathbb{Z}[x, y, z]$$

by expressing a polynomial $P(x; y, z)$ in terms of x_A in the first factor and x_B in the second factor. That is, if

$$P(x; y, z) = \sum_i c_i(y, z) P_i(x_A) Q_i(x_B)$$

then set

$$\mathrm{sp}_A(P) = \sum_i c_i(y, z) P_i \otimes Q_i.$$

We define the splitting coefficients $d_w^{u,v}(A \mid y; z)$ by the expansion

$$\mathfrak{S}_w(x; y) = \sum_{u,v} d_w^{u,v}(A \mid y; z)\, \mathfrak{S}_u(x_A; y)\, \mathfrak{S}_v(x_B; z)$$

In the ordinary Schubert polynomial case ($y = z = 0$), these splitting coefficients are often called coproduct coefficients in the literature. They have a geometric interpretation as the coefficients of the map on cohomology induced by the embedding $Fl(V) \times Fl(W) \hookrightarrow Fl(V \oplus W)$ [2]. For double Schubert polynomials with the same coefficient variables, an analogous construction in equivariant cohomology appears in [1].

We will express the splitting coefficients explicitly in terms of triple structure constants. To begin, we give a definition:

Definition 2.1.2 (μ_A)**.** Let μ be a dominant permutation, and let A be a set of positive integers (finite or infinite). Indexing A in increasing order and writing it as a function on the corresponding integer interval, we then define μ_A to be the dominant permutation such that

$$\mathfrak{c}_i^*(\mu_A) = \mathfrak{c}_{A(i)}^*(\mu)$$

where we define $\mathfrak{c}_i^*(\mu_A) = 0$ if $i > |A|$.

Theorem 2.1 (Product–splitting equivalence)**.** *Suppose $w \in S_\infty$ and let μ be a dominant permutation such that $\ell(w\mu) = \ell(\mu) - \ell(w)$. Let A be a set of positive integers and let $B = \mathbb{N} \setminus A$. Then*

$$\mathfrak{S}_w(x; y) = \sum_{\substack{u:\ell(u\mu_A)=\ell(\mu_A)-\ell(u) \\ v:\ell(v\mu_B)=\ell(\mu_B)-\ell(v)}} c^{w\mu}_{u\mu_A, v\mu_B}(-y; -z)\, \mathfrak{S}_u(x_A; y)\, \mathfrak{S}_v(x_B; z)$$

That is to say, fixing μ, we have that $d_w^{u,v}(A \mid y; z) = 0$ unless $\ell(u\mu_A) = \ell(\mu_A) - \ell(u)$ and $\ell(v\mu_B) = \ell(\mu_B) - \ell(v)$, in which case

$$d_w^{u,v}(A \mid y; z) = c^{w\mu}_{u\mu_A, v\mu_B}(-y; -z)$$

Remark. Theorem 2.1 is new even for ordinary Schubert polynomials.

Example 2.1.3. We provide a simple example. Set $A = \{1\}$, $B = \{2\}$. We have

$$\begin{aligned}\mathfrak{S}_{[1,4,2,3]}(x; y) = (z_1 - y_2)(z_1 - y_3) + (z_1 - y_3)\, \mathfrak{S}_{[2,1]}(x_2; y) + (z_1 + z_2 - y_2 - y_3)\, \mathfrak{S}_{[2,1]}(x_1; z) \\ + \mathfrak{S}_{[2,1]}(x_2; y)\, \mathfrak{S}_{[2,1]}(x_1; z) + \mathfrak{S}_{[3,1,2]}(x_2; y) + \mathfrak{S}_{[3,1,2]}(x_1; z)\end{aligned}$$

For the dominant permutation, we can choose $\mu = [\![2, 2]\!]$. Then $w\mu = [\![1, 1]\!]$. We have $\mu_A = \mu_B = [\![1, 1]\!]$. To find the first coefficient, we compute

$$c^{[\![1,1]\!]}_{[\![1,1]\!],[\![1,1]\!]}(-y; -z) = (z_1 - y_2)(z_1 - y_3)$$

For the second, $u = [2, 1]$, and $u\mu_A = [\![0, 1]\!]$.

$$c^{[\![1,1]\!]}_{[\![0,1]\!],[\![1,1]\!]}(-y; -z) = z_1 - y_3$$

For the third term, $v = [2, 1]$, so

$$c^{[\![1,1]\!]}_{[\![1,1]\!],[\![0,1]\!]}(-y; -z) = z_1 + z_2 - y_2 - y_3$$

For the fourth term, we have

$$c^{[\![1,1]\!]}_{[\![0,1]\!],[\![0,1]\!]}(-y; -z) = 1$$

For the fifth term, $u = [3, 1, 2]$, so $u\mu_A$ is the identity, so that the coefficient is 1, and similarly for the last term where $v = [3, 1, 2]$.

The reader might have noticed that the right-hand side has z variables, but the left-hand side does not. In fact, all of the z variables cancel. Nevertheless, the splitting coefficients are polynomials in the differences $z_i - y_j$ with nonnegative integer coefficients, as established below.

The product–splitting equivalence also transfers positivity results to splitting coefficients. We first address the equivariant case $y = z$, whose Graham positivity was proved in [1].

Corollary 2.1.4 (Positivity of double Schubert polynomial splitting coefficients)**.** *Let $w \in S_\infty$, let $A, B \subseteq \mathbb{N}$ be complementary and write*

$$\mathfrak{S}_w(x; y) = \sum_{u,v} d_w^{u,v}(A \mid y; y)\, \mathfrak{S}_u(x_A; y)\, \mathfrak{S}_v(x_B; y)$$

Then the coefficient $d_w^{u,v}(A \mid y; y)$ is a polynomial in the positive roots $y_i - y_{i+1}$ with nonnegative integer coefficients.

Proof. We have that

$$d_w^{u,v}(A \mid y; y) = c^{w\mu}_{u\mu_A, v\mu_B}(-y; -y)$$

for some μ, since we can always pick some dominant permutation μ satisfying $\ell(w\mu) = \ell(\mu) - \ell(w)$. By the main result of [10], $c^{w\mu}_{u\mu_A, v\mu_B}(y; y)$ is a polynomial in the negative roots $y_{i+1} - y_i$ with nonnegative integer coefficients. Substitution of $-y$ into this gives the result. □

Corollary 2.1.5 (Positivity of mixed-variable double Schubert polynomial splitting coefficients)**.** *Let $w \in S_\infty$, let $A, B \subseteq \mathbb{N}$ be complementary and write*

$$\mathfrak{S}_w(x;y) = \sum_{u,v} d_w^{u,v}(A \mid y;z)\,\mathfrak{S}_u(x_A;y)\,\mathfrak{S}_v(x_B;z)$$

Then the coefficient $d_w^{u,v}(A \mid y;z)$ is a polynomial in the terms $z_i - y_j$ with nonnegative integer coefficients.

Proof. By Theorem 2.1, each nonzero splitting coefficient has the form $c^{w\mu}_{u\mu_A, v\mu_B}(-y;-z)$ for a suitable dominant permutation μ. Gao and Xiong's positivity theorem [9, Theorem 1.1] states that $c^{w\mu}_{u\mu_A, v\mu_B}(y;z)$ is a polynomial in the differences $y_i - z_j$ with nonnegative integer coefficients. Substituting $(y,z) \mapsto (-y,-z)$ sends each $y_i - z_j$ to $z_j - y_i$, proving the claim. □

Surprisingly, the proof of Theorem 2.1 is essentially a simple algebraic observation derived from the Cauchy formula. The Cauchy formula for double Schubert polynomials is well-known [17], and related analysis in the context of nil-Coxeter algebra and the generating function for Schubert polynomials can be found in [7]. Given the general importance of the following theorem in our proof of the product–splitting equivalence, we note that it was proved for the universal double Schubert polynomials defined by Fulton [8] by Kirillov in [12]. The result then descends immediately to the case of double Schubert polynomials. For completeness, we provide here a direct elementary proof for double Schubert polynomials since it is all that is needed.

Proposition 2.1.6 (The generalized Cauchy formula for double Schubert polynomials)**.** *Let $w \in S_\infty$ be a permutation and consider an additional infinite set of variables $a = \{a_i \mid i \geq 1\}$. Then we have*

$$\mathfrak{S}_w(x;y) = \sum_{\substack{w_1 w_2 = w \\ \ell(w_1)+\ell(w_2)=\ell(w)}} \mathfrak{S}_{w_2}(x;a)\,\mathfrak{S}_{w_1}(a;y)$$

Proof. In the polynomial ring $\mathbb{Z}[x,y,a]$ we have that the expression of the polynomial $\mathfrak{S}_w(x;y)$ in the basis of double Schubert polynomials $\mathfrak{S}_u(x;a)$ as a module over $\mathbb{Z}[y,a]$ can be written as

$$\mathfrak{S}_w(x;y) = \sum_{u \in S_\infty} \partial^u(\mathfrak{S}_w(x;y))|_{x=a}\,\mathfrak{S}_u(x;a)$$

We have that

$$\partial^u(\mathfrak{S}_w(x;y))|_{x=a} = \begin{cases} \mathfrak{S}_{wu^{-1}}(a;y) & \text{if } \ell(wu^{-1}) = \ell(w) - \ell(u) \\ 0 & \text{otherwise.} \end{cases}$$

Thus

$$\mathfrak{S}_w(x;y) = \sum_{\substack{u \in S_\infty \\ \ell(wu^{-1})=\ell(w)-\ell(u)}} \mathfrak{S}_{wu^{-1}}(a;y)\,\mathfrak{S}_u(x;a)$$

This is clearly equivalent to the result. □

We will also need the following factorization property for double Schubert polynomials associated with dominant permutations.

Lemma 2.1.7. *Suppose A and B are complementary sets of positive integers, and let μ be dominant. Then*

$$\mathfrak{S}_\mu(x;y) = \mathfrak{S}_{\mu_A}(x;y_A)\,\mathfrak{S}_{\mu_B}(x;y_B)$$

Proof. Let $A = \{a_1 < a_2 < \cdots\}$ and $B = \{b_1 < b_2 < \cdots\}$. Since A and B are complementary, we have that

$$\mathfrak{S}_\mu(x;y) = \prod_{i=1}^{\infty} E_{\mathfrak{c}^*_{a_i}(\mu)}(x;y_{a_i}) \prod_{j=1}^{\infty} E_{\mathfrak{c}^*_{b_j}(\mu)}(x;y_{b_j})$$

By definition of μ_A and μ_B, and the characterization of double Schubert polynomials corresponding to dominant permutations as products of factorial elementary symmetric polynomials in [22, Lemma 4.5], the result follows. □

Proof of Theorem 2.1. By Lemma 2.1.7, we have

$$\mathfrak{S}_\mu(x;y) = \mathfrak{S}_{\mu_A}(x;y_A)\,\mathfrak{S}_{\mu_B}(x;y_B)$$

Let a and b be additional sets of variables. Combining this identity with the Cauchy formula, we obtain

$$\mathfrak{S}_\mu(x;y) = \sum_{u,v} \mathfrak{S}_{u\mu_A}(x;a)\,\mathfrak{S}_{v\mu_B}(x;b)\,\mathfrak{S}_{u^{-1}}(a;y_A)\,\mathfrak{S}_{v^{-1}}(b;y_B)$$

Multiplying this out, we obtain

$$\begin{aligned}\mathfrak{S}_\mu(x;y) &= \sum_w \sum_{u,v} \mathfrak{S}_{w\mu}(x;a) c^{w\mu}_{u\mu_A,v\mu_B}(a;b)\,\mathfrak{S}_{u^{-1}}(a;y_A)\,\mathfrak{S}_{v^{-1}}(b;y_B)\\ &= \sum_w \sum_{u,v} \mathfrak{S}_{w\mu}(x;a) c^{w\mu}_{u\mu_A,v\mu_B}(a;b)\,\mathfrak{S}_{u}(-y_A;-a)\,\mathfrak{S}_{v}(-y_B;-b)\\ &= \sum_w \mathfrak{S}_{w\mu}(x;a)\,\mathfrak{S}_w(-y;-a)\end{aligned}$$

Therefore, substituting gives

$$\mathfrak{S}_w(x;y) = \sum_{u,v} c^{w\mu}_{u\mu_A,v\mu_B}(-y;-z)\,\mathfrak{S}_u(x_A;y)\,\mathfrak{S}_v(x_B;z)$$

as desired. □

To justify calling this an "equivalence" theorem, we need to show that splitting coefficients can be obtained from product coefficients in this way.

Definition 2.1.8 (The left dominant approximation $\mathfrak{d}^*(u)$, $\mathfrak{d}^*(u,v)$)**.** To any permutation u we associate a dominant permutation $\mathfrak{d}^*(u)$, the *left dominant approximation of* u, as follows. If u is dominant, set $u = \mathfrak{d}^*(u)$. Otherwise, if i is the maximal index such that $\mathfrak{c}^*_i(u) < \mathfrak{c}^*_{i+1}(u)$, set $\mathfrak{d}^*(u) = \mathfrak{d}^*(s_i u)$. Note that we have

$$\mathfrak{d}^*(u) = \mathfrak{d}(u^{-1})^{-1}$$

Then we define $\mathfrak{d}^*(u,v)$ to be the dominant permutation such that

$$\mathfrak{c}(\mathfrak{d}^*(u,v)) = \mathfrak{c}(\mathfrak{d}^*(u)) + \mathfrak{c}(\mathfrak{d}^*(v))$$

Lemma 2.1.9. *Suppose $u, v \in S_\infty$. Then there exist complementary subsets of positive integers A, B such that*

$$\mathfrak{d}^*(u,v)_A = \mathfrak{d}^*(u)$$

and

$$\mathfrak{d}^*(u,v)_B = \mathfrak{d}^*(v)$$

Proof. This can be deduced from Lemma 2.1.7 and Lemma 1.2.5. □

Theorem 2.2 (Every product coefficient occurs as a splitting coefficient)**.** *Suppose $u, v, w \in S_\infty$. Let A and B be subsets of $\mathbb{N}$ as in Lemma 2.1.9. Then $c^w_{u,v}(y;z) = 0$ unless $\ell(w\mathfrak{d}^*(u,v)^{-1}) = \ell(\mathfrak{d}^*(u,v)) - \ell(w)$. In that case, we have that*

$$c^w_{u,v}(y;z) = d^{u\mathfrak{d}^*(u)^{-1}, v\mathfrak{d}^*(v)^{-1}}_{w\mathfrak{d}^*(u,v)^{-1}}(A \mid -y;-z)$$

Proof. Consider the product

$$\mathfrak{S}_{\mathfrak{d}^*(u,v)}(x;y) = \mathfrak{S}_{\mathfrak{d}^*(u)}(x;y_A)\,\mathfrak{S}_{\mathfrak{d}^*(v)}(x;y_B)$$

Introduce new sets of variables a, b, and write

$$\mathfrak{S}_{\mathfrak{d}^*(u)}(x;y_A)\,\mathfrak{S}_{\mathfrak{d}^*(v)}(x;y_B) = \sum_{u',v'} \mathfrak{S}_{u'}(x;a)\,\mathfrak{S}_{v'}(x;b)\,\mathfrak{S}_{u'\mathfrak{d}^*(u)^{-1}}(-y_A;-a)\,\mathfrak{S}_{v'\mathfrak{d}^*(v)^{-1}}(-y_B;-b)$$

In particular, the coefficient of $\mathfrak{S}_{u\mathfrak{d}^*(u)^{-1}}(-y_A;-a)\,\mathfrak{S}_{v\mathfrak{d}^*(v)^{-1}}(-y_B;-b)$ is

$$\mathfrak{S}_u(x;a)\,\mathfrak{S}_v(x;b) = \sum_w c^w_{u,v}(a;b)\,\mathfrak{S}_w(x;a)$$

Considering the Cauchy formula on $\mathfrak{S}_{\mathfrak{d}^*(u,v)}(x;y)$, we obtain

$$\sum_{w':\ell(w'\mathfrak{d}^*(u,v)^{-1})=\ell(\mathfrak{d}^*(u,v))-\ell(w')} \mathfrak{S}_{w'}(x;a)\,\mathfrak{S}_{w'\mathfrak{d}^*(u,v)^{-1}}(-y;-a)$$

By linear independence of double Schubert polynomials, if $c^w_{u,v}(a;b) \neq 0$ then $\mathfrak{S}_w(x;a)$ must occur as a term in this formula. Thus, $\ell(w\mathfrak{d}^*(u,v)^{-1}) = \ell(\mathfrak{d}^*(u,v)) - \ell(w)$. By applying Theorem 2.1, we obtain that

$$c^w_{u,v}(y;z) = c^{(w\mathfrak{d}^*(u,v)^{-1})\mathfrak{d}^*(u,v)}_{(u\mathfrak{d}^*(u)^{-1})\mathfrak{d}^*(u,v)_A,(v\mathfrak{d}^*(v)^{-1})\mathfrak{d}^*(u,v)_B}(y;z) = d^{u\mathfrak{d}^*(u)^{-1},v\mathfrak{d}^*(v)^{-1}}_{w\mathfrak{d}^*(u,v)^{-1}}(A \mid -y;-z)$$

And we have the result. □

The following result illustrates the equivalence between products and splitting coefficients for permutations with a given number of descents.

Theorem 2.3 (Descent product–splitting equivalence)**.** *Let K be a set of descents. If w has descents contained in K, then there exists a set of descents K' with $|K| = |K'|$ and a $w' \in S_\infty$ with descents contained in K' as well as $A \subseteq \mathbb{N}$ such that for all $u, v \in S_\infty$ with descents contained in K there exist $u', v' \in S_\infty$ such that*

$$c^w_{u,v}(y;z) = d^{u',v'}_{w'}(A \mid y;z)$$

Conversely, if K' is a set of descents, $w' \in S_\infty$ has descents contained in K', and $A \subseteq \mathbb{N}$, then there exists a set of descents K with $|K| \leq |K'|$ and a $w \in S_\infty$ with descents contained in K such that for all $u', v' \in S_\infty$ there exist $u, v \in S_\infty$ with descents contained in K such that the above equality holds.

Proof. If $w \in S_n$, let μ_1 be the longest element of S_n^K, and let μ be such that

$$\mathfrak{c}_i(\mu) = 2\mathfrak{c}_i(\mu_1)$$

for all i. Then there exist A, B such that $\mu_A = \mu_B = \mu_1$. By Theorem 2.1, for any $u, v \in S_n^{\mathbb{N}\setminus K}$ (the set of minimal length coset representatives of the parabolic subgroup generated by s_i for $i \notin K$), we have that

$$c^w_{u,v}(y;z) = d^{u\mu_1^{-1},v\mu_1^{-1}}_{w\mu^{-1}}(A \mid y;z)$$

Let K' be the descent set of μ^{-1}. Then $|K'| = |K|$, and we may select $w' = w\mu^{-1}$, $u' = u\mu_1^{-1}$, and $v' = v\mu_1^{-1}$. Since μ^{-1} has descents contained in K', so does $w\mu^{-1}$. This covers all u and v with descents contained in K such that $c^w_{u,v}(y;z) \neq 0$ since such u and v must be contained in S_n^K.

Conversely, suppose w' has descents contained in K'. Let $\mu = \mathfrak{d}^*(w)^{-1}$, and let K be the descent set of μ. Since w has at most $|K'|$ descents, μ has at most $|K'|$ descents, so $|K| \leq |K'|$. Then for any A and $B = \mathbb{N} \setminus A$, μ_A and μ_B have descents contained in K. For any u', v' we have that

$$d^{u',v'}_{w'}(A \mid y;z) = c^{w'\mu}_{u'\mu_A,v'\mu_B}(y;z)$$

by Theorem 2.1. All three indices in the product coefficient have descents contained in K. Set $w = w'\mu$, $u = u'\mu_A$, and $v = v'\mu_B$, and the result follows. □

Thus, a combinatorial formula for products in k-step flag varieties is equivalent to a formula for splitting coefficients of Schubert polynomials indexed by permutations with k descents. In fact, known results imply combinatorial positivity of splitting coefficients of double Schubert polynomials indexed by permutations with at most two descents, and of ordinary Schubert polynomials with at most three descents.

Corollary 2.1.10. *The puzzle rule in* [5] *computes $d^{u,v}_w(A \mid y;y)$ Graham-positively for all u, v, w, and A such that w has at most two descents.*

Corollary 2.1.11. *The puzzle rule proved in* [13] *computes $d^{u,v}_w(A \mid 0;0)$ positively for all u, v, w, and A such that w has at most three descents.*

2.2. **A historical missed opportunity: separated descents for ordinary Schubert polynomials.** Let $k \geq 1$ be an integer, and consider the variable-splitting expansion

$$\mathfrak{S}_w(x) = \sum_{u,v} d_w^{u,v}(k)\, \mathfrak{S}_u(x_1, \ldots, x_k)\, \mathfrak{S}_v(x_{k+1}, \ldots, x_n)$$

The coefficients $d_w^{u,v}(k)$ were computed in 1991 by Macdonald in [17, (4.19)] and shown to be nonnegative. This variable-splitting expansion is often called the Schubert polynomial coproduct in the literature. The condition of separated descents was introduced in 2019 in [14]. Recall that permutations u, v are said to have separated descents if there exists a $p > 0$ such that $\ell(us_i) > \ell(u)$ for all $i < p$ and $\ell(vs_i) > \ell(v)$ for all $i > p$. We now interpret this condition in terms of Theorem 2.2. We begin with the following proposition.

Proposition 2.2.1. *Let $u \in S_\infty$ and suppose $j > 0$ is an integer. Then $\ell(us_j) < \ell(u)$ if and only if $\ell(\mathfrak{d}^*(u)s_j) < \ell(\mathfrak{d}^*(u))$.*

Proof. We prove this by induction on $\ell(\mathfrak{d}^*(u)) - \ell(u)$, the result being clear if $u = \mathfrak{d}^*(u)$. If u is not dominant, let i be the maximal index such that $\mathfrak{c}_i^*(u) < \mathfrak{c}_{i+1}^*(u)$. We know by definition that $\mathfrak{d}^*(u) = \mathfrak{d}^*(s_i u)$. If $\ell(us_j) < \ell(u)$, then $\ell(s_i us_j) < \ell(s_i u)$. By the induction hypothesis, $\ell(\mathfrak{d}^*(u)s_j) < \ell(\mathfrak{d}^*(u))$.

Conversely, suppose $\ell(us_j) > \ell(u)$ and assume the induction hypothesis. We claim that $\ell(s_i us_j) > \ell(s_i u)$. If instead we assume that $\ell(s_i us_j) < \ell(s_i u)$, then in fact $s_i u = us_j$ by the exchange property. This means that $u^{-1}(i) = j$, $u^{-1}(i+1) = j+1$. However, this implies that $\mathfrak{c}_i^*(u) = \mathfrak{c}_{i+1}^*(u)$, contradicting the choice of i. Thus we can conclude that $\ell(s_i us_j) > \ell(s_i u)$, and hence by the induction hypothesis $\ell(\mathfrak{d}^*(u)s_j) > \ell(\mathfrak{d}^*(u))$, as desired. This completes the induction and proves the proposition. □

If u and v have separated descents, therefore, so do $\mathfrak{d}^*(u)$ and $\mathfrak{d}^*(v)$. We also have the following.

Lemma 2.2.2. *Let μ be a dominant permutation, let $A = [k]$ for some $k > 0$, let $B = \mathbb{N} \setminus [k]$, and let $p = \mathfrak{c}_k^*(\mu)$. Then $\ell(\mu_A s_i) > \ell(\mu_A)$ for all $i < p$ and $\ell(\mu_B s_i) > \ell(\mu_B)$ for all $i > p$, or in other words μ_A and μ_B have separated descents.*

Proof. If $p = 0$, then $\mu_B = 1$, and the result follows immediately. Otherwise, observe that the elements of $\mathfrak{c}^*(\mu_A)$ are the right descents of μ_A, and similarly for $\mathfrak{c}^*(\mu_B)$. Since $\mathfrak{c}^*(\mu)$ is a partition, μ_A has no descents i such that $i < p$ because $\mathfrak{c}_j^*(\mu_A) \geq p$ for all j, and μ_B has no descents i such that $i > p$ because $\mathfrak{c}_j^*(\mu_B) \leq p$ for all j. This establishes the claim. □

Conversely, suppose $\ell(us_i) > \ell(u)$ for all $i < p$, $\ell(vs_i) > \ell(v)$ for all $i > p$, and $\mathfrak{d}^*(u,v)_A = \mathfrak{d}^*(u)$ and $\mathfrak{d}^*(u,v)_B = \mathfrak{d}^*(v)$, then A and B can be chosen to be intervals (one initial and finite, and the other complementary and infinite). Therefore, setting $y = z = 0$, to compute $c_{u,v}^w(0;0)$ from Theorem 2.2, the coefficients we need are exactly those in Macdonald's coproduct. Thus, a formula for separated descents could already have been obtained in 1991, 28 years before its appearance in [14].

We now establish positivity of these splitting coefficients in the equivariant and mixed-variable settings.

Theorem 2.4 (Separated splitting theorem)**.** *Let $k > 0$ and let $A = [k]$ be the initial interval. Then the coefficients*

$$d_w^{u,v}\,(A \mid y; z)$$

appearing in the expansion

$$\mathfrak{S}_w(x; y) \;=\; \sum_{u,v} d_w^{u,v}\,(A \mid y; z)\; \mathfrak{S}_u(x_1, \ldots, x_k; y)\; \mathfrak{S}_v(x_{k+1}, \ldots, x_n; z),$$

*where n denotes the last descent of w, have the following properties: 1. They are polynomials with nonnegative coefficients in the differences $z_i - y_j$. 2. After the specialization $z = y$, they are *Graham positive*—that is, polynomials in the simple roots $y_i - y_{i+1}$ with nonnegative integer coefficients. Moreover, nonnegativity in the variables $z_i - y_j$ also holds when $A = \mathbb{N} \setminus [k]$.*

Proof. Assume first that $A = [k]$ and set $B = \mathbb{N}\setminus[k]$. We have that $d_w^{u,v}\,(A \mid y; z) = c_{u\mu_A, v\mu_B}^{w\mu}(-y; -z)$ for some dominant permutation μ. There exists a p such that $\ell(u\mu_A s_i) > \ell(u\mu_A)$ for all $i < p$ and $\ell(v\mu_B s_i) > \ell(v\mu_B)$ for all $i > p$ by Lemma 2.2.2. Hence, [22, Theorem 3.1] allows us to compute this coefficient nonnegatively as a polynomial in $z_i - y_j$ that is manifestly Graham positive after the substitution $z = y$. For $A = \mathbb{N} \setminus [k]$, the opposite descent conditions hold, and Theorem 1.4 applies to provide positivity as a polynomial in $z_i - y_j$. When we set $z = y$, the coefficients become equal, but Graham positivity is directly visible (without cancellation) only when we use [22, Theorem 3.1] to compute the coefficient. □

2.3. **Reduction formulas for product coefficients by varying the dominant permutation.** Theorem 2.1 does not depend on which dominant permutation is chosen to define the splitting coefficient. Suppose we wish to compute the product coefficient $c_{u,v}^{w}(y;z)$. Then, by Theorem 2.2, we have that

$$c_{u,v}^{w}(y;z) = d_{w\mathfrak{d}^*(u,v)^{-1}}^{u\mathfrak{d}^*(u)^{-1},v\mathfrak{d}^*(v)^{-1}}(A \mid -y;-z)$$

Choosing $\mathfrak{d}^*(u,v)$ as the dominant permutation in which to transform this into a splitting coefficient may, however, be overkill. Write

$$\widehat{w} = w\mathfrak{d}^*(u,v)^{-1}$$
$$\widehat{u} = u\mathfrak{d}^*(u)^{-1}$$
$$\widehat{v} = v\mathfrak{d}^*(v)^{-1}$$

We may think of $\widehat{u}$, $\widehat{v}$, and $\widehat{w}$ as the "base indices" of this splitting coefficient. We have that

$$c_{u,v}^{w}(y;z) = c_{\widehat{u}\mathfrak{d}^*(u),\widehat{v}\mathfrak{d}^*(v)}^{\widehat{w}\mathfrak{d}^*(u,v)}(y;z)$$

Choose

$$\mu = \mathfrak{d}(\widehat{w}^{-1})$$

and choose sets of positive integers A and B such that

$$\mathfrak{d}^*(u,v)_A = \mathfrak{d}^*(u)$$

and

$$\mathfrak{d}^*(u,v)_B = \mathfrak{d}^*(v)$$

Then we may transform this to obtain

$$c_{u,v}^{w}(y;z) = c_{\widehat{u}\mu_A,\widehat{v}\mu_B}^{\widehat{w}\mu}(y;z)$$

It is always the case that $\ell(\widehat{w}\mu) \leq \ell(w)$, so this transformation never increases the length of the upper index. At worst, $\mathfrak{d}(\widehat{w}^{-1}) = \mathfrak{d}^*(u,v)$, so the indices remain unchanged. The following example demonstrates that this method can convert a coefficient that we do not immediately know how to compute into one that we do know how to compute.

Example 2.3.1. Let

$$u = [\![1,0,2]\!]$$
$$v = [\![0,2,1]\!]$$
$$w = [\![4,0,1]\!]$$

Then

$$c_{u,v}^{w}(y;z) = y_1 + y_5 - z_1 - z_2$$

None of the known formulas allows a direct positive computation of this coefficient. The reduction method, however, transforms it into $c_{[\![0,0,2]\!],[\![0,1,1]\!]}^{[\![0,1,2]\!]}(y;z)$, which the reader may verify is equal to the original coefficient. Since

$$\mathfrak{S}_{[\![0,1,1]\!]}(x;z) = E_{2,3}(x;z)$$

this case can be computed using the Pieri formula [22, Theorem 7.1]. Alternatively, one may apply the factorial Schur product formula [18, Theorem 3.1], as the expression is a product of factorial Schur polynomials.

$$\begin{aligned} s_{(2)}(x_1,x_2,x_3;y)s_{(1,1)}(x_1,x_2,x_2;z) = {} & ((y_1-z_1)(y_5-z_2)+(y_2-z_1)(y_1-z_1)+(y_2-z_2)(y_5-z_2))s_{(2)}(x_1,x_2,x_3;y) \\ & + (y_1+y_2-z_1-z_2)s_{(3)}(x_1,x_2,x_3;y) + (y_1+y_5-z_1-z_2)s_{(2,1)}(x_1,x_2,x_3;y) \\ & + s_{(3,1)}(x_1,x_2,x_3;y) + s_{(2,1,1)}(x_1,x_2,x_3;y) \end{aligned}$$

whereas the original product we were trying to compute, $\mathfrak{S}_{[\![1,0,2]\!]}(x;y)\,\mathfrak{S}_{[\![0,2,1]\!]}(x;z)$, has 25 nonzero double Schubert polynomial terms.

The `schubmult` Python package can display product and splitting coefficients in positive form. For the most general algorithms applicable to arbitrary polynomials, obtaining such a presentation is computationally intractable-even in small cases-when approached directly through integer programming without problem-specific optimizations. By contrast, when the reduction formulas introduced in this paper are applied, together with other positive formulas established in prior work, the package frequently produces positive presentations for cases of reasonable size. This statement reflects empirical evidence rather than a quantified result. Notably, such optimizations were essential to render positivity testing of coefficients feasible in the case $n = 6$.

Acknowledgments

We thank Kellen Meyers for assistance with large-scale computations used to test our conjectures. We are also grateful to Frank Sottile for many valuable discussions; his perspective, especially as much of our work generalizes his, has been invaluable.

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